

A Review on Machine Learning approaches for assessing Urban Heat Islands.

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Abstract

Urban Heat Island (UHI) is a phenomenon where the temperatures in the core of the cities by aggravated by upto 8 °C in comparison to the periphery. These elevated ambient temperatures pose significant challenges to urban systems, living conditions, and exacerbate air pollution in urban areas. Over the years, the use of Machine Learning (ML) techniques to study the UHI effect has noticeably increased. This study aims to systematically review the role of ML in addressing this crucial phenomenon and its potential in analysing the UHI effect.

Our findings indicate that the use of ML for the processing of satellite data has experienced a steady growth over the past two decades, with a notable increase between 2018 and 2021. The most common application of ML was found to be the measurement of Land Surface Temperature (LST) in quantifying the UHI effect in conjunction with variables like building energy consumption, air temperature and wind movement. However, there is room for improvement and increased utilization of ML, particularly in multi-temporal investigations and predictive modelling. The key challenges that must be addressed include ensuring data accessibility, data pre-processing, complexity of the UHI phenomenon, lack of ground truth data for validation and increasing awareness of ML among stakeholders. The integration of ML techniques offers substantial potential for enhancing our comprehension of UHI dynamics and informing targeted mitigation efforts in urban environments.

Keywords – Machine Learning; Urban Heat Island; Land Surface Temperature; Remote sensing; UHI mitigation; Data acquisition

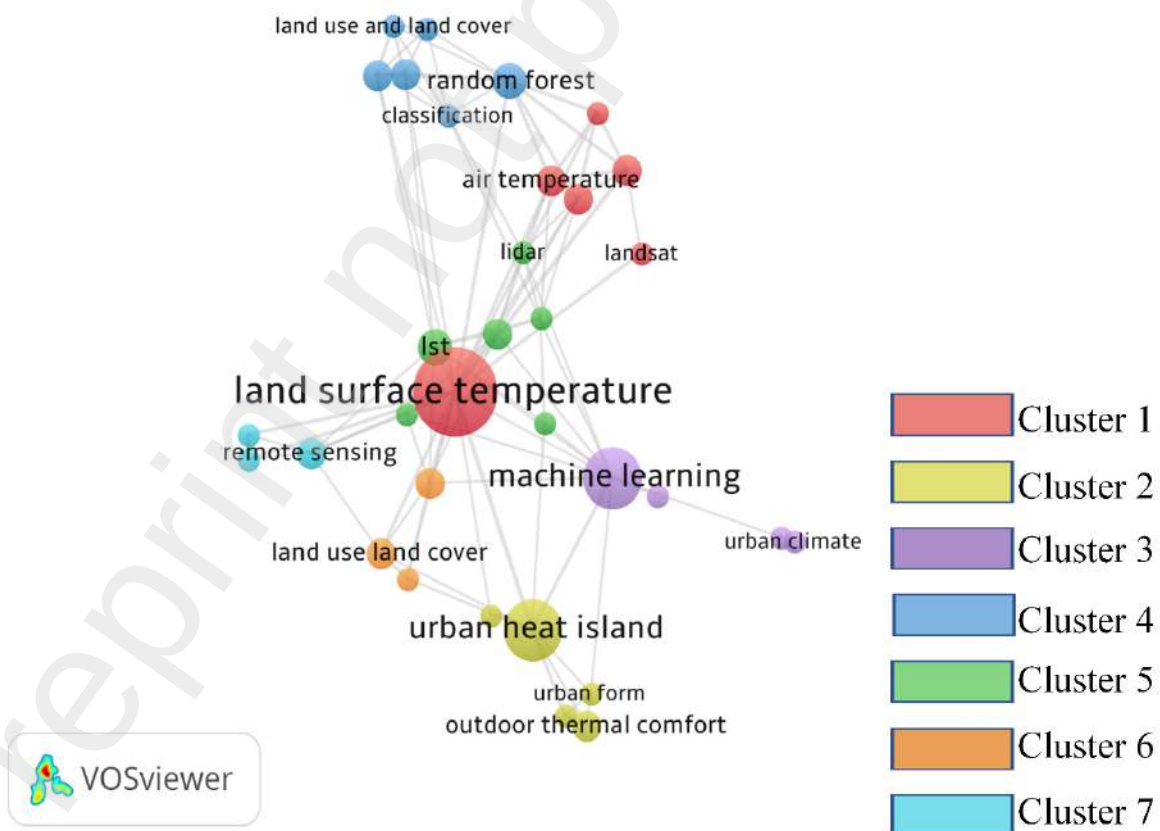
1 Introduction

The urban climate is significantly affected by economic growth and urbanization, resulting in increased urban heating and the Urban Heat Island (UHI) effect. UHI, the most commonly used indicator of urban climate change, refers to the rise in air temperature in metropolitan areas compared to surrounding rural and suburban regions due to anthropogenic heat release, limited green spaces, and reduced air circulation in urban canyons [1,2].

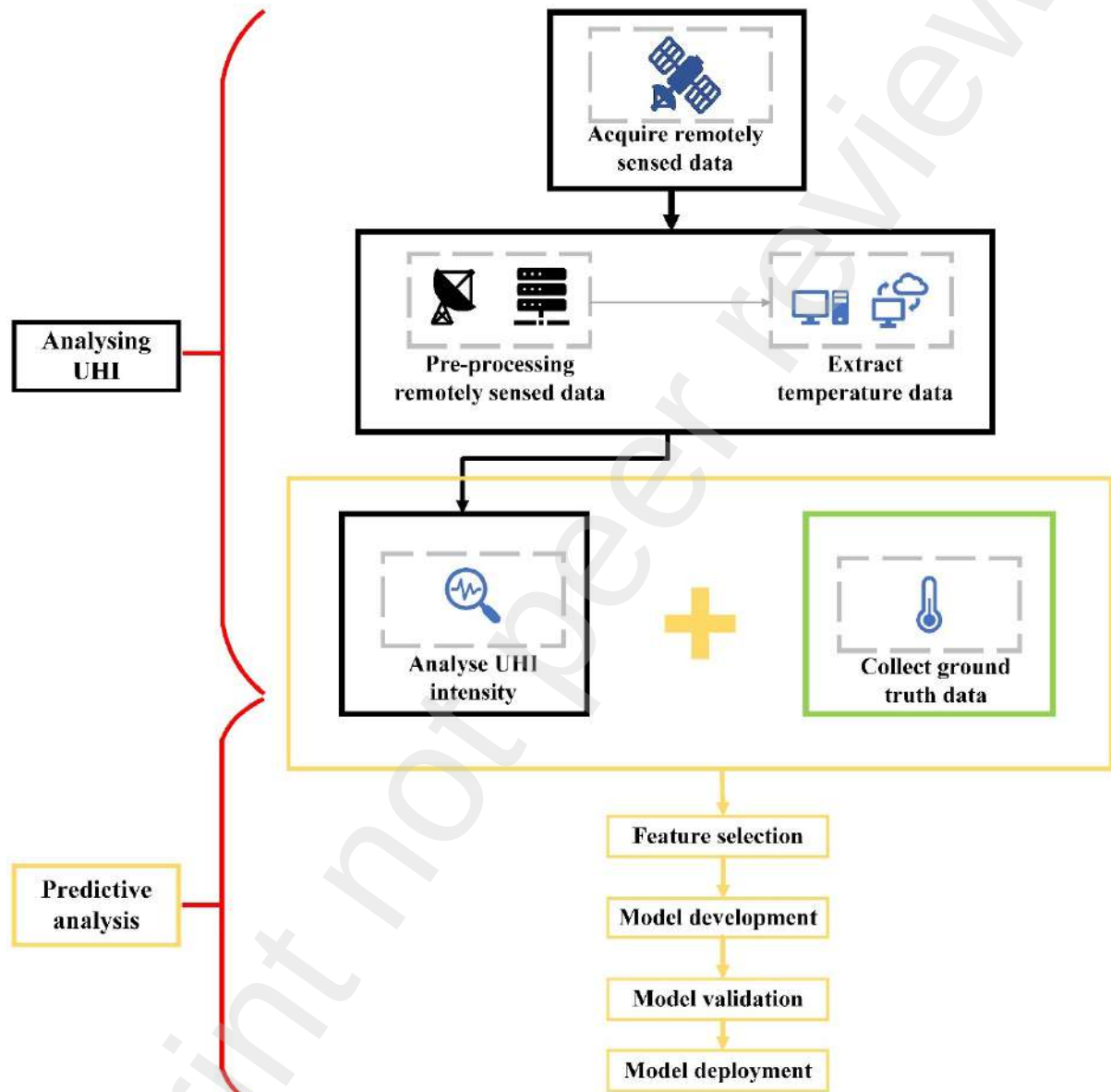
UHI has implications on energy demand, consumption, and balances in buildings, as well as human health and thermal comfort. The use of air conditioning systems increases with higher temperatures in urban areas, resulting in higher energy consumption, exacerbate peak electricity demand and greenhouse gas emissions, negatively impacting energy sustainability [3–6]. According to Santamouris, in Athens, UHI has nearly doubled building cooling demand and nearly tripled peak power consumption, resulting in a reduction of the Coefficient of Performance (COP) of mechanical cooling systems by up to 25% [3,7]. Furthermore, Akbari et al. have found that in US cities with a population greater than 100,000, peak electrical load will increase by 1.5–2% for every 1°F increase in temperature [8]. UHI also alters energy balances in buildings and affects district heating and cooling, creating a strain on urban systems [9]. This makes UHI a critical issue for the built environment in terms of energy sustainability, resilience, and climate adaptability. Additionally, high air temperatures caused by UHIs and extreme heat events can lead to increased mortality and morbidity, infrastructure failure, and a decline in quality of life. Socially and economically vulnerable populations are particularly susceptible to the impacts of heat.

Recent research has resulted in the development of various methodologies for analysing the UHI effect and proposing mitigation measures to counterbalance it. The launch of numerous earth observation satellites, such as Landsat, Moderate Resolution Imaging Spectroradiometer

(MODIS) and European Centre for Medium-Range Weather Forecasts (ECMWF) v5 has increased the availability of thermal data for researchers to analyse, model, and predict temperatures at any location on earth. The most popular thermal variable analysed in UHI assessments is the land surface temperature (LST). Figure 1 shows a bibliometric term map depicting the occurrence of keywords in recent publications resulting from a keyword search of "UHI", "Urban Heat Island" and "Machine Learning" on ScienceDirect from 2018 to 2022. Additionally, the development of various thermal indices, such as Urban Thermal Field Variance Index (UTFVI), Surface Urban Heat Island Index (SUHII), UHI Ratio Index (URI) and Urban Heat Island Index (UHII), has made the quantification of the UHI effect increasingly feasible. The availability of large global datasets has led stakeholders in UHI to rely less on traditional modeling, statistical techniques, and expert knowledge. Instead, they are now focusing on the use of Machine Learning (ML) to analyze these datasets.



38 Figure 1 Bibliometric term map showing the co-occurrence of frequently used keywords in 7
 39 clusters, represented by different colors in the legend. Circle size indicates keyword
 40 occurrence, while line thickness represents inter-cluster co-occurrence.



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42 Figure 2 Flowchart depicting the process of UHI assessment and predictive analysis using
 43 remote sensing and machine learning. The steps involve acquiring and pre-processing remote
 44 sensing data, segmenting and classifying the data using machine learning algorithms, and

deriving urban environmental parameters. Statistical and spatial modeling techniques are then used to identify key drivers of UHI formation and develop predictive models.

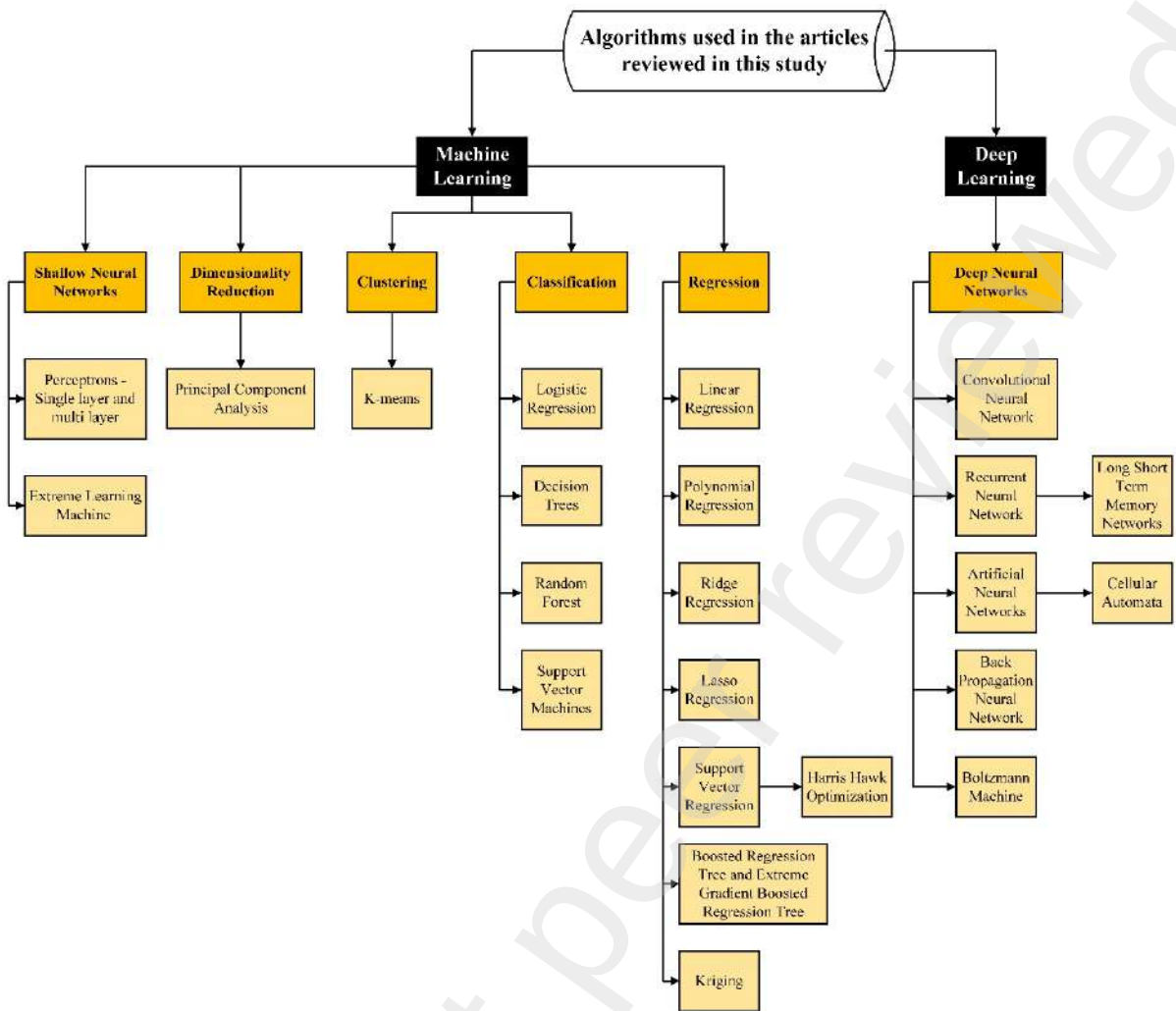
1.1 Structure of the review

In structuring this review, we have conscientiously organized the narrative to reflect a more methodical and coherent presentation of UHI studies. Recognizing the imperative to transcend a mere enumeration of ML methodologies, we have opted for a thematic organization centered around key perspectives of UHI research. Specifically, the manuscript now elucidates UHI calculation methodologies, UHI prediction models, and investigations into influential factors contributing to UHI phenomena like urban growth. This deliberate framework aims to elevate the scholarly merit of the review, aligning it with the exacting standards expected in top-tier scientific journals. By adopting this thematic arrangement, our objective is to offer a nuanced and comprehensive exploration of UHI dynamics, thereby enhancing the academic rigor and impact of this contribution to the scientific discourse.

Within the realm of UHI assessment, the integration of Machine Learning (ML) manifests a transformative potential [10]. This section navigates through diverse applications of ML, specifically in the enhancement of Land Surface Temperature (LST) retrieval, analysis, prediction, and forecasting from remote sensing data. Emphasizing the importance of methodological precision, thermal image processing for LST retrieval encompasses the estimation of thermal radiance through statistical or physical modeling approaches. ML algorithms, such as support vector regression (SVR), artificial neural networks (ANN), and random forest (RF), proficiently model these processes, ensuring high accuracy in correlating LST with other remote sensing-based indices. Supervised and unsupervised ML techniques, including clustering and classification, are instrumental in deciphering patterns and extracting valuable information from retrieved LST data. Notably, convolutional neural networks (CNN)

69 rooted in deep learning (DL) exhibit promising capabilities in capturing intricate spatial and
70 temporal dependencies within LST data, outperforming traditional methods. In the context of
71 LST prediction and forecasting, recurrent neural networks (RNNs) and long short-term
72 memory (LSTM) models, specializing in time-series analysis, navigate the complex temporal
73 dynamics of LST data. Proficient in capturing both short-term and long-term trends, these
74 models emerge as indispensable tools for predicting heat waves and extreme heat events.
75 Figure 2 provides a visual representation, outlining the sequential stages of UHI assessment
76 and predictive analysis, illustrating the pervasive influence of ML across the entire process.

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79 *Figure 3 A flowchart that categorizes the machine learning algorithms (ML algorithms) used*
 80 *in the papers reviewed in this study. The study aimed to analyze the use of ML algorithms in a*
 81 *particular domain and included a comprehensive search of relevant literature. The chart also*
 82 *includes subcategories, such as decision trees, clustering, and neural networks, among others,*
 83 *to provide a detailed overview of the ML algorithms used in the study.*

84 **1.2 Objectives of this review**

85 The primary objective of this review is to provide a comprehensive overview of ML approaches
 86 used for the assessment of UHIs. It can be divided into three sub-objectives:

1. To identify and analyze the various types of ML algorithms utilized for the assessment of UHIs, including their strengths and limitations.
2. To evaluate the performance of these approaches in accurately predicting UHIs and its associated factors.
3. To provide insights into the most effective ML techniques and the data requirements necessary for an accurate UHI assessment.

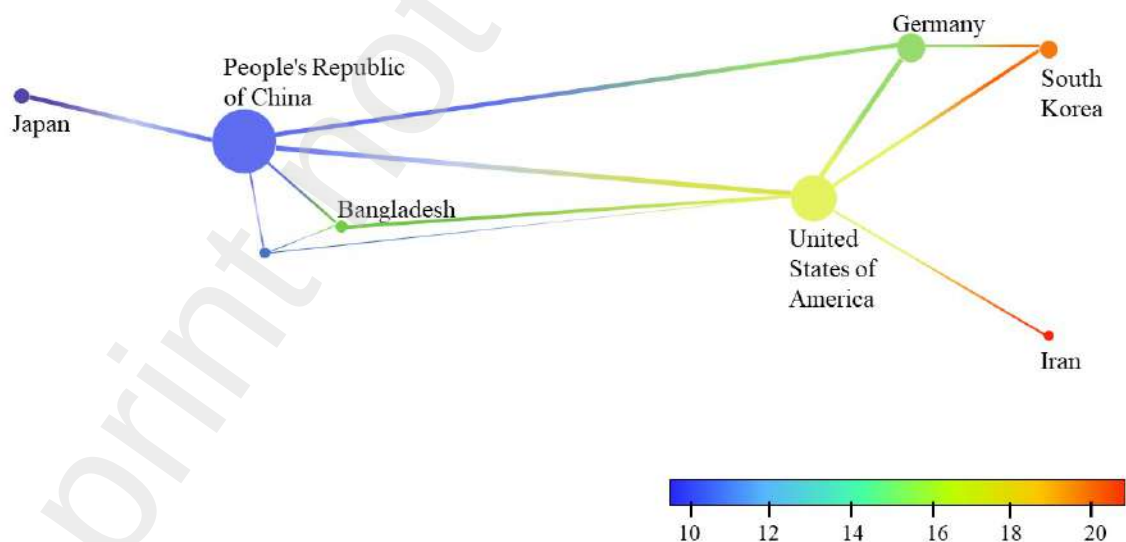
Ultimately, this review aims to provide a valuable resource for researchers and practitioners involved in UHI assessment, offering an overview of the latest ML approaches and their potential applications in this field.

1.3 Papers reviewed

To fulfil the above stated objectives, Scopus, Web of Science, and ScienceDirect were consulted to evaluate the implementation of ML techniques in UHI evaluation. The search phrases "Machine Learning" , "Urban Heat Island" and "UHI" were used, yielding 2,168 articles on October 1, 2022. After deleting the duplicates and applying a time filter, 93 articles that contained both keywords were selected for further examination. These articles were then programmatically filtered based on the prevalence of ML-related phrases in their titles yielding a total of 68, as shown in Figure 5. Figure 4 shows the cross country citation analysis done in VOSViewer for the selected articles.

We keep our focus on papers published between 2001 to 2021. As demonstrated in Figure 6, ML methods gained popularity after 2015-2016, with an exponential increase in publications related to UHI assessment. Additionally, we conducted a yearly distribution analysis of publications to determine the trends in research focus. Our analysis revealed a significant increase in the number of published papers from 2018 to 2022. These findings suggest that there has been a shift in research focus towards the application of ML techniques for UHI in

111 recent years. The selection of articles for inclusion in this review was driven by a discerning
 112 focus on the intricate relationship between Urban Heat Islands (UHIs) and Land Surface
 113 Temperature (LST). It is noteworthy that the examination of pseudo indicators, particularly
 114 Land Use and Land Cover (LULC), has been a prevalent and insightful approach in the study
 115 of UHI dynamics. Many studies deploy Machine Learning (ML) methods not solely for their
 116 intrinsic capabilities but, rather, as instrumental tools in the investigation of LULC. This
 117 nuanced approach is underpinned by the acknowledged correlation between LULC and both
 118 LST and UHI effects. Consequently, a substantial portion of the reviewed literature pertains to
 119 the application of ML methodologies for land use and cover classification, recognizing their
 120 direct relevance to the understanding of UHI dynamics. While the emphasis extends beyond
 121 UHI per se, this deliberate focus on LULC is crucial for elucidating the broader environmental
 122 factors influencing UHI effects and contributes to the holistic comprehension of urban thermal
 123 regimes.



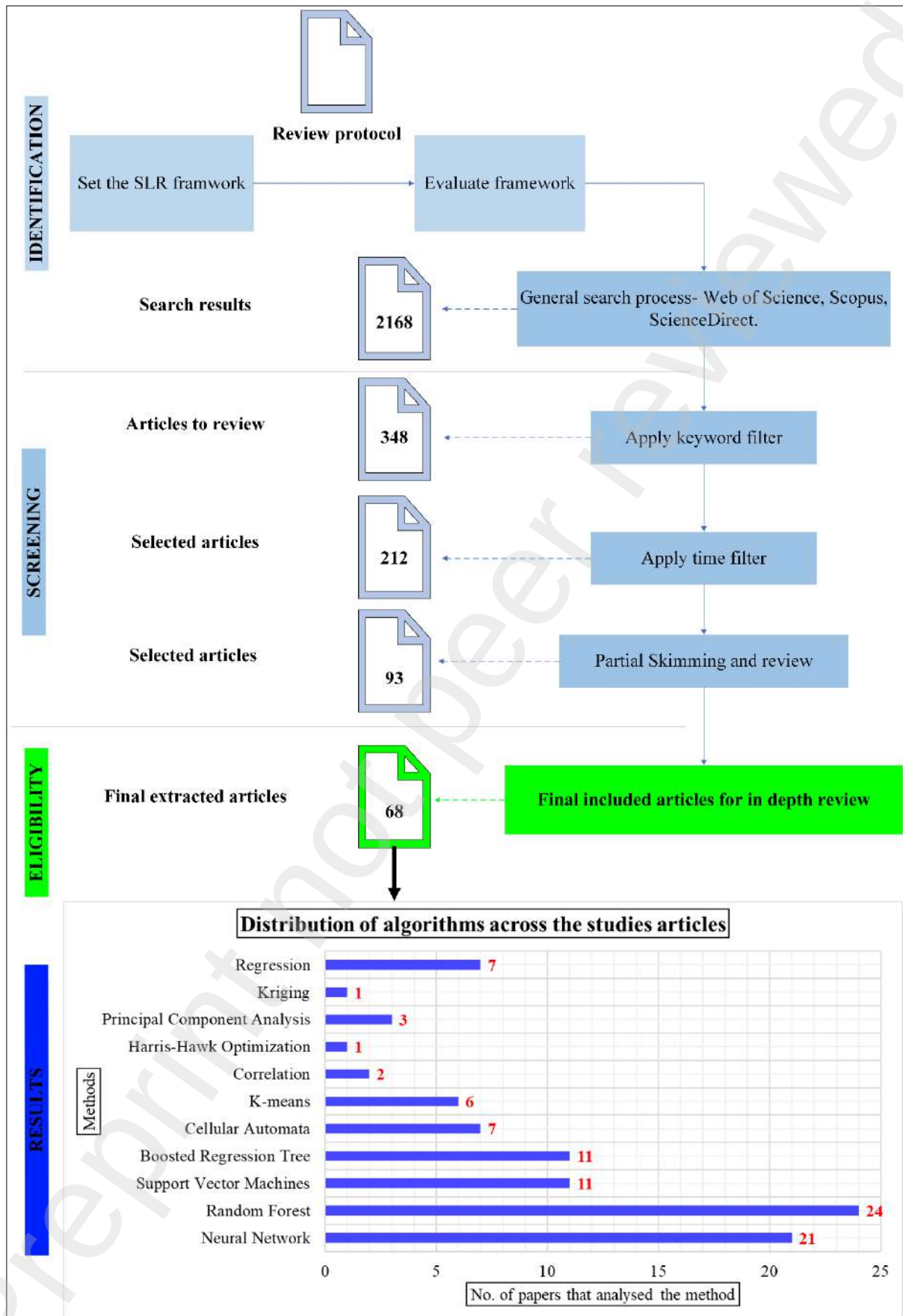
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125 *Figure 4 Country-wise citations in the field of ML in UHI - Network map representing the*
 126 *strength based on publications and citations in 7 countries. Bigger is the circle, more is the*

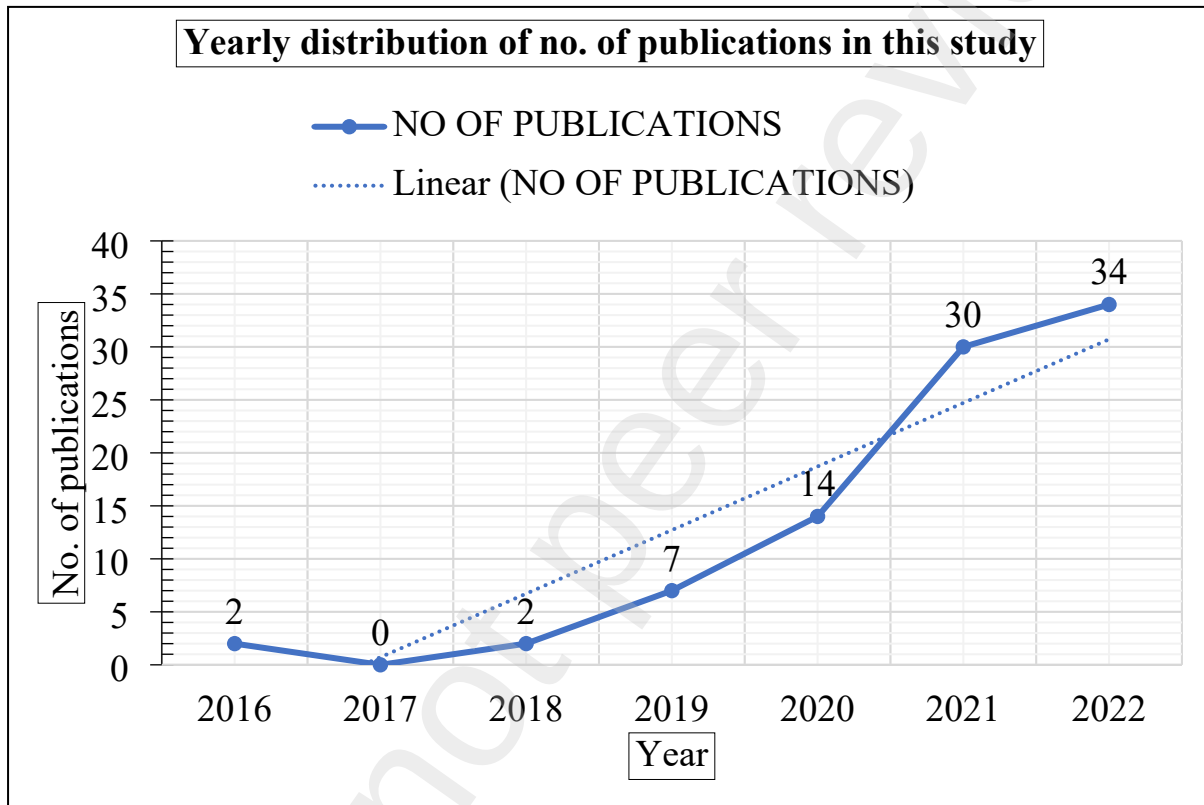
127 *number of publications, and the thickness of the connecting lines represents the strength of*
128 *cross-country citation between the two countries. The color denotes the number of publications*
129 *cited in other countries as per the scale.*

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133 *Figure 5 Methodology for selection of articles and distribution of ML algorithms. The figure*
 134 *demonstrates the process of selecting 68 articles from the research database using keyword*
 135 *filtering and other selection criteria. The bar plot in the figure presents the distribution of ML*
 136 *algorithms covered in the selected articles, including the number of articles that employed each*
 137 *algorithm.*



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139 *Figure 6 Yearly distribution of publications utilized in the Study. The line chart depicts the*
 140 *number of publications utilized in this study, organized by year. The data represents the*
 141 *distribution of publications over the specified time period and highlights the trends in the*
 142 *increasing use of ML in UHI studies.*

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2 Using ML in UHI assessment

The following section of this review will discuss the use of ML approaches in the assessment of UHIs, with a focus on meeting the previously stated objectives. By analyzing the strengths and limitations of various ML algorithms, we aim to provide insights into the most effective techniques for accurately predicting UHIs and their associated factors like LULC, LST, UTFVI, SUHII, UHII, URI etc. Through the evaluation of performance metrics, including accuracy, sensitivity, and specificity, we will assess the effectiveness of these approaches in addressing the challenges of UHI assessment. Additionally, this section will examine the data requirements necessary for accurate assessment of UHIs, including the type, source and quality of data required for different ML approaches. By synthesizing the previously stated objectives, we aim to position this review as a valuable resource for researchers and practitioners involved in the assessment of UHIs, offering insights into the latest ML approaches and their potential applications in this field. In this section of the paper, we will provide a detailed analysis of the papers organized according to the ML algorithms used in the analysis.

2.1 Neural Networks

Neural networks are a promising approach for UHI assessment, used for predictive analysis, thermal indices estimation, synthetic data creation and simulating UHI-related variables. This section will discuss their contributions to meeting the review objectives. Papers are categorized by the type of neural network used to provide a detailed analysis of their strengths and limitations for addressing UHI assessment challenges.

2.1.1 *Artificial Neural Network (ANN)*

Artificial Neural Networks (ANNs) have emerged as a popular approach for predictive analysis in UHI assessment. In a study by Mohammad et al., ANNs were utilized alongside multi-date

Landsat data to forecast LULC and seasonal LST over Ahmedabad, India. The findings projected significant increases in urban built-up areas by 2025 and 2030, accompanied by corresponding decreases in farmland. This shift is expected to result in elevated LST levels, particularly during summer and winter seasons [11]. Similarly, researchers focused on cities in Bangladesh, employing ANNs to predict LULC changes, seasonal LST and UTFVI over a two-decade period. Their results indicated notable expansions of urban areas at the expense of green cover, highlighting ongoing urbanization trends in the region[12–14].

Aldousari et al. conducted a comprehensive analysis of LULC changes in Kuwait, utilizing ANNs to forecast future scenarios. Their study revealed substantial conversions of bare land and vegetation into built-up areas, leading to considerable increases in mean LST and UHI values over the years. Projections for 2031 suggested further declines in vegetation cover and significant temperature rises [15].

In a separate study by Maniatis et al., ANNs were employed to estimate human heat perception based on the Physiological Equivalent Temperature (PET). The researchers found that by using only air temperature and humidity measurements obtained with high spatial resolution across a mountainous region as input variables, they could estimate complex thermal indices. The simulated PET index using validated ANN models was used to estimate the best season to visit the Ainos National Park on Kefalonia Island, Greece [16].

Moreover, ANNs have been instrumental in simulating and quantifying the UHI effect. Symonds et al. presented the design of a nationwide model aimed at simulating indoor overheating and air pollution. This model took into account building stock, climate change, outdoor air pollution levels, and occupant behavior, based on EnergyPlus simulations. The study also conducted a metamodeling analysis of the risks of overheating and air pollution in the English dwelling stock. During the model creation process, the performance of NNs was

compared to that of support vector regression (SVR), with NNs demonstrating a performance improvement of over 50% compared to SVR [17].

ML techniques were also employed to simulate and quantify the impact of industrial activities on LST levels to minimize extreme UHI (E-UHI) effects. The findings indicated that the size of all types of industries affects both factory LST (LST_f) and surrounding LST (LST_s), while the geometry of steel factories primarily affects LST_s and LST [18].

2.1.2 Recurrent Neural Network (RNN)

The application of RNNs represents a paradigm shift in our ability to model and predict complex spatio-temporal dynamics. RNNs, a class of artificial neural networks designed to capture sequential dependencies in data, offer unprecedented capabilities in analyzing temporal variations of UHI phenomena. By leveraging the recurrent connections within their architecture, RNNs excel in capturing temporal patterns and dependencies inherent in UHI datasets, ranging from hourly temperature fluctuations to long-term trends. This unique capability enables RNNs to uncover intricate relationships between various environmental factors and UHI intensity, facilitating more accurate and robust predictions of urban heat dynamics. RNNs have mainly been used for simulation of UHI related variables like tropospheric O₃ and other air pollutants. Feng et al. analyse air pollution in Hangzhou, China using RNN and RF for 9 million citizens and the upcoming Asian Games. The study found that ML models are more efficient, accurate, and cost-effective in simulating all air pollutants without the need for a pollution inventory. The RF-generated Feature Importance shows that carbon monoxide, nitrogen dioxide, and particulate matter are the main contributors to air pollution. The study also found that RNN outperforms other ML models in simulating tropospheric ozone, which is highly tied to time series data. RNN demonstrates rapid convergence, short calculation time, adaptability to forecast episodes, low memory

requirements, higher accuracy, and lower cost compared to traditional atmospheric models. The authors show that a 10-fold-trained RNN outperforms the widely used chemical reaction mechanism CB05 implemented in Community Multiscale Air Quality Modeling System (CMAQ), suggesting that traditional atmospheric models may be improved through the use of ML [19].

2.1.3 Convolutional Neural Network (CNN)

CNNs have been used in modelling Urban Evapo-Transpiration (UET), another variable that serves as a pseudo-indicator of UHI. Vulova et al. evaluated their method using a two-year dataset from two flux towers in Berlin, Germany. The study tested the integration of remote sensing and geographic information system (GIS) data into models using four predictor scenarios and compared the performance of CNNs and RF. The best-performing models were employed to predict UET values for 2019. The use of GIS data generated from flow footprints improved model accuracy, particularly in cases where meteorological data was limited. The best-performing scenario, which utilized both meteorological and GIS data, had an root mean square error (RMSE) of 0.0239 mm/h with RF and an R^2 of 0.840 with a 1D CNN for the more vegetated area [20].

2.1.4 Back Propagation Neural Network (BPNN)

Back Propagation Neural Networks (BPNNs) offer a potent tool for modeling and forecasting UHI dynamics. BPNNs, a foundational architecture in artificial neural networks, excel in learning complex nonlinear relationships inherent in UHI datasets. By iteratively adjusting network weights through backpropagation of errors, BPNNs effectively capture the intricate interactions between various environmental factors and UHI intensity. Their ability to generalize from training data to unseen instances enables BPNNs to provide accurate predictions of UHI behavior. X. Zhang et al. conducted a study in Changxing, a small Chinese

city, where they collected data from multiple sources and developed an ML approach using a BPNN to estimate urban building cooling energy (UBCE) functions based on building energy form (BEF). UBCE is directly related to UHI. The study found that BEF has an impact on the density, function, and form of UBCE. The BPNN was able to make predictions of UBCE functions with a mean absolute percentage error (MAPE) of 10-20%. The results of the predictions indicated that the average UBCE functions varied greatly [21].

2.1.5 Extreme Learning Machine (ELM)

Extreme Learning Machines (ELMs) emerge as a formidable technique for uncovering intricate patterns and relationships within UHI datasets. ELMs, characterized by their single-layer feedforward neural network architecture, offer rapid learning capabilities and superior generalization performance. By randomly initializing input weights and analytically determining output weights, ELMs efficiently process large-scale UHI data, yielding accurate predictions of UHI intensity and spatial distribution. ELMs have mainly been used to create synthetic data using data fusion and enhancing the accuracy of ML models. Bai et al. provided a unique data fusion approach for obtaining LST datasets at 30m spatial resolution using daily MODIS and Landsat Enhanced Thematic Mapper Plus (ETM+) images. The resolution of the Landsat ETM+ thermal infrared (TIR) data was reduced from 60m to 30m using an ELM method and neural network regression model. Then, the Spatio-temporal Adaptive Data Fusion Algorithm for Temperature Mapping (SADFAT) was used to fuse the MODIS LST and upgraded Landsat ETM+ TIR data, generating high-resolution synthetic data. The synthetic data was evaluated for both testing and satellite simulations [22]. In another study, ELM is a single-hidden-layer feed-forward neural network learning technique that uses 50,000 hidden nodes to enhance accuracy. The ML approach identifies input-to-output mappings. The results showed that except for ELM, the other three ML models captured all instances, including the failures of the Weather Research and Forecasting (WRF) and CMAQ models in late January.

ELM's sole permissible month was January, and it performed poorly in the following three months. The linear learning model included ELM, and the results indicated that the MLP outperformed ELM, indicating that ozone generation is nonlinear [19].

2.1.6 Markov Chain Model (MCM)

Markov Chain Model (MCM) emerges as a potent tool for exploring temporal dynamics and transitions within urban thermal environments. Leveraging the principles of stochastic processes, MCMs enable the modeling of UHI and its psuedoindicators evolution over time, capturing the probabilistic transitions between different UHI states. By characterizing the underlying dynamics of temperature changes and spatial patterns, MCMs offer valuable insights into the long-term trends and persistence of UHI phenomena. The study by Ozturk utilized the Markov chain model to analyse the spatiotemporal distribution of LULC and make predictions about future changes. The hybrid model was constructed using the Land Change Modeler (LCM) in Idrisi TerrSet. This model combined the Markov chain with Cellular Automata(CA) methods to forecast LULC change. The Markov chain component of the model predicts the transition of a cell from one state to another, but it does not take into account the impact of neighbouring cells. This lack of spatial modelling is a limitation of the Markov chain component of the hybrid model [23].

2.1.7 Boltzmann Machine

Applying weak supervision and thought-based learning to an improved restricted Boltzmann machine (RBM) model, this research developed an intensity level identifier for a UHI. The identification process begins with automatic annotation and successively optimizes the model parameters until the target identifier is complete. With limited information on the weak labelling of the target training sample, the algorithm can still build a geographical distribution map of UHI intensity [24].

2.1.8 *Deep Neural Network (DNN)*

DNNs have mainly been used in estimating the benefits of UHI mitigation measures. Huang et al. reconstructed a trained neural network model with a validation accuracy of 94.17% to estimate the removal of pollutants by dry deposition from green roofs in Shijiazhuang, China. According to their results, the implementation of green roofs in the city could remove 1,210,465,106 kg/year of pollutants. This study provides a valuable foundation for environmental policy formulation and urban green system development [25]. Li et al. used a deep forest (DF) model to estimate ozone concentrations at a 300 m resolution in China, incorporating Landsat 8 IR bands and meteorological data. The DF model outperformed other ML algorithms, such as generalized regression neural regression networks and RF, in terms of accuracy and mapping. Cross-validation methods, including sample-based, station-based, time-based, and extrapolation validation, demonstrated the consistency of DF estimates with station data, with determination coefficients ranging from 0.68 to 0.93 [26].

2.1.9 *Fusion of black box and white box models*

In a study, both white-box and black-box approaches were used to simulate urban heating. White-box models are physically based and simple but require expertise to operate, while black-box models are easier to use but are less transparent. The study reviewed the use of these two approaches in UHI prediction and discussed the benefits of combining them for accurate UHI prediction and mitigation during the design and operation phases of urban areas. The physical-based white-box models are reliable for UHI evaluation, but complex white-box models may lead to incorrect conclusions if not handled correctly. On the other hand, black-box models, though opaque, need fewer calculations. The study concluded that combining the two approaches can improve UHI forecasting and mitigation [27].

2.2 Random Forest (RF)

Random Forest (RF), a versatile machine learning algorithm, is extensively employed in UHI assessment due to its efficacy in handling complex relationships among variables. Its application spans various domains within UHI research, including land use land cover (LULC) classification, predictive analysis, UHI estimation, synthetic data creation, UHI mitigation, thermal indices estimation, and air temperature simulation.

In LULC classification, RF demonstrates remarkable performance, particularly in identifying areas with high impervious surface coverage, a significant contributor to UHI [28]. For instance, studies by Kulithalai et al. and Karunaratne et al. utilized RF alongside other machine learning techniques to accurately classify LULC types in different regions [29,30]. Similarly, studies by Tepanosyan et al. and Logan et al. leveraged RF classifiers to distinguish between various land cover types and assess their impact on temperature variations [31,32].

Moreover, RF models have been employed for predictive analysis in UHI assessment, enabling the estimation of UHI intensity based on environmental and anthropogenic factors. Vulova et al. and Schneider utilized RF regression models to predict urban air temperature with high precision, incorporating a range of predictors to enhance model accuracy [20,33].

RF also finds application in UHI mitigation strategies by identifying areas suitable for urban greening and tree planting, one of the most popular mitigation strategy against the UHI effect. Helletsgruber et al. utilized RF regression analysis to assess the impact of tree species on surface cooling, demonstrating the efficacy of RF in quantifying the cooling effects of vegetation cover [34].

Additionally, RF has been instrumental in UHI estimation based on remotely sensed data, allowing for the prediction of UHI intensity using thermal infrared imagery. ArcGIS-based studies such as ArcUHI integrated RF regression models to accurately estimate UHI intensity over different time periods, aiding in the development of targeted mitigation measures [35].

Synthetic data creation using RF has also been explored, as demonstrated by Xu et al., who utilized RF-based strategies for downscaling LST data, enhancing spatial resolution for improved UHI assessment [36]. Furthermore, RF has been utilized in the estimation and comparison of thermal indices, such as the Universal Thermal Climate Index (UTCI), providing insights into human heat sensitivity and outdoor thermal comfort variations throughout different seasons [37].

Finally, RF-based simulations of air temperature play a crucial role in UHI assessment, aiding in the evaluation of UHI effects on human health and the development of mitigation strategies. Chen et al. and Yoshikazu et al. employed RF models to simulate air temperature accurately, particularly during heatwaves or under different urbanization patterns [38,39].

The collective body of research underscores the versatility and effectiveness of RF in various facets of UHI assessment, ranging from LULC classification to predictive modeling and UHI mitigation strategies..

2.3 Support Vector Machine (SVM)

One of the primary applications of Support Vector Machines (SVM) in UHI assessment lies in LULC classification. SVM techniques have demonstrated efficacy in discerning various land types, including vegetation, water bodies, and impervious surfaces, which are crucial for identifying areas with heightened imperviousness contributing to UHI. In a study by Cetin et al., mean shift clustering facilitated the segmentation of high vegetation tree tops, enabling classifiers with 25 spatial and intensity-based criteria to differentiate between deciduous and

coniferous tree species within metropolitan landscapes [40]. Similarly, Kulithalai et al. compared SVM performance with other classification techniques, revealing an accuracy of 82.5%, slightly lower than RF (89.0%) but comparable to CART (87.5%). Despite SVM's competitiveness, misclassifications near water bodies were observed, especially during dry seasons, where RF showcased superior accuracy [29].

Furthermore, SVM finds application in estimating UHI-related variables like solar radiation and thermal indices, offering insights into areas susceptible to UHI formation. Guermoui et al. introduced a novel approach for daily global horizontal solar radiation estimation by combining two SVM models. The first model predicts solar radiation, while the second quantifies prediction errors, resulting in enhanced performance compared to conventional SVM models [41]. Similarly, Bakhashwain utilized SVM to estimate global solar radiation in Sharurha, training the model with air temperature and humidity data. The study exhibited the potential of SVM in accurately estimating solar radiation levels [42].

In addition to LULC classification and variable estimation, SVM has also been utilized in UHI prediction and forecasting scenarios. A study by Tepanosyan et al. employed SVM and ANN to analyze and predict LULC changes over two decades. Results indicated significant alterations in land cover, corresponding with escalated UHI values and pronounced impacts on Urban Thermal Fractional Vegetation Index (UTFVI) zones [14]. This integration of SVM in predictive models offers crucial insights for understanding UHI dynamics and facilitating informed urban planning strategies [31].

2.4 Boosted Regression Tree (BRT)

One of the prominent applications of Boosted Regression Trees (BRT) in UHI assessment lies in predictive analysis. BRT models are adept at forecasting UHI intensity based on diverse environmental and anthropogenic factors, encompassing air temperature, humidity, and urban

morphology. This capability facilitates the identification of UHI-prone areas and the formulation of effective mitigation strategies. Balogun et al. showcased gradient boosting as one of the most accurate ML algorithms for predicting air pollutant concentration, emphasizing the sensitivity of air quality to climate change and the critical role of urban planning interventions in combating urban air pollution [43]. Similarly, a study conducted by Grunwald et al. utilized BRTs to predict cold-air channels in German cities, highlighting their significance in mitigating UHI effects by introducing cooler air from rural areas or inner-city parks into densely populated metropolitan regions [44].

In addition to predictive analysis, BRT has been instrumental in estimating LST and Heat Flux, leveraging remotely sensed data like thermal infrared imagery. Schneider et al. demonstrated the efficacy of BRT models in estimating LST by incorporating factors such as temperature, vegetation index, and solar zenith angle, thereby offering valuable insights for UHI assessment and mitigation efforts [33]. Furthermore, Sun et al. utilized BRT analysis to investigate the impact of 3D urban architecture on LST, revealing the importance of both 2D and 3D architectural metrics in influencing surface temperature in urban environments [45]. Moreover, Alavipanah et al. employed BRT analysis to discern the influence of 2D and 3D urban indicators on surface temperature, providing a novel approach for urban planners to identify elements that affect surface temperature in metropolitan regions [46].

Furthermore, BRT has been integral in UHI mitigation strategies by identifying areas with high potential for urban greening. Amindin et al. found that NDVI, land use, and elevation significantly influenced surface temperature, with vegetation along streets and highways contributing to cooler LSTs near major routes [47]. This highlights the importance of employing BRT models in urban planning to identify areas for green infrastructure development, thereby mitigating UHI effects and enhancing urban livability.

2.5 Cellular Automata (CA)

Cellular Automata (CA) is a mathematical and computational modeling approach that has been applied in various fields, including UHI assessment. CA has shown promise in UHI assessment due to its ability to simulate the complex and dynamic nature of urban growth.

One common use of CA in UHI assessment is in combination with ANNs for predictive analysis. CA-ANN models can be used to predict UHI intensity and spatial distribution based on various environmental and anthropogenic factors, such as land use, land cover, and population density as discussed in the ANNs subsection section above. An analysis of Landsat 4-5 TM and 8 OLI images from 1995 to 2020 was conducted to evaluate the historical condition of LULC and UTFVI and to predict future changes for 2025 and 2030. The results anticipate a 42% and 44% expansion in urban built-up areas between 2025 and 2035, accompanied by losses in green cover (21% and 22%), bare land (20% and 21%), and water bodies (1%) [48].

In addition to predictive analysis, CA has also been used for the simulation of urban growth. CA models can simulate the expansion of urban areas, which can help identify areas that are at high risk of UHI formation in the future. The future urban growth scenario can be simulated using the CA model and an ANN learning process. The MOLUSCE plugin for QGIS 2.18 was utilized, which includes a six-step prediction algorithm [14].

2.6 K-means

K-means clustering has emerged as a promising approach in UHI assessment, owing to its capacity to group similar data points into clusters, thereby offering insights into the spatio-temporal distribution of UHI phenomena. Kwak et al. conducted a study employing LST analysis alongside k-means clustering to investigate regional UHI trends. They identified that factors such as green and blue space ratios, transportation networks, and housing distributions

influence UHI patterns and intensities [49]. Furthermore, in determining UHI intensity, LST inversion, K-RBM clustering, and an enhanced RBM identification strategy were found crucial. The updated RBM identification process exhibited the highest classification accuracy with a Kappa coefficient of 0.8861, outperforming traditional k-means clustering due to its susceptibility to local optimization issues [16].

Another common application of K-means in UHI assessment involves analyzing city growth patterns. Tao et al. utilized k-means clustering to identify areas experiencing significant urban expansion, aiding in pinpointing regions prone to UHI formation. Their study revealed that urban growth was more pronounced during the day and positively correlated with a city's population level, indicating a spatial evolution trend from east to west between 2005 and 2018 [22]

Moreover, K-means has proven instrumental in generating synthetic data for UHI studies. Chang et al. developed the Urban Microscale Temperature Forecast (UMTF) model, incorporating k-means clustering alongside SVM techniques and global ensembles. Utilizing geomorphometry datasets, the UMTF model downscaled temperature extremes, enhancing resolution across Hong Kong [50].

In addition to synthetic data generation, K-means has been pivotal in clustering based on temperature data. Fadel et al. employed k-means clustering as an unsupervised ML approach to classify grid tiles by zone properties. Their findings underscored the utility of k-means clustering in revealing city-specific classes or Land Cover Zone (LCZ) transition types, complementing GIS-LCZ classification [51]. Furthermore, k-means clustering facilitated group creation to examine the relationship between urbanization and temperature rise, supporting results obtained from Cluster Development Index (CDI) analysis. The study

highlighted that built area growth in housing complexes contributes to heat island formation, as indicated by CDI and cluster analysis [52].

2.7 Correlation

Correlation analysis has shown promise in UHI assessment due to its ability to measure the strength and direction of the relationship between two variables. One common use of correlation analysis in UHI assessment is in studying thermal indices. Thermal indices are commonly used to estimate the thermal environment of an urban area, and correlation analysis can be used to identify the variables that have the greatest impact on thermal indices. According to Darbani et al., the PET with urban form factors. The H/W ratio has an effect on the PET index rate: as the H/W ratio increases, the PET index rate lowers, leading to improved OTC. The average mean radiant temperature (MRT), which measures the amount of reflection from walls, rooftops, and open areas, also affects OTC [53].

Correlation analysis has also been used in studying the dispersion of LST in urban areas. LST is a critical variable in UHI assessment, and correlation analysis can be used to identify the variables that have the greatest impact on LST dispersion. A study examines the dispersion of LST and spectral markers. The researchers found that vegetative land influences urban expansion and creates a negative correlation between the NDVI and soil adjusted vegetation index (SAVI) ($R^2 = 0.20$ and 0.15 , respectively). On the other hand, urbanization and industrialization cause a positive link between LST, the Normalised Difference Built-Up Index (NDBI), and the urban index (UI) ($R^2=0.61$ and 0.27). The Modified Normal Difference Water Index (MNDWI) and the normalized difference bare soil index (NDBal) have low correlation values (0.0003 and 0.04 , respectively). The researchers concluded that urbanization, climate change, and global warming have raised the temperature in Kolkata and nearby cities, leading

to a growing UHI due to human activities. This analysis helps identify recent temperature changes and supports the development of a sustainable urban layout [54].

2.8 Harris Hawk Optimization

According to Kurniawan et al., their study aims to predict ozone concentration in 14 subdistricts of JABODETABEK using Harris Hawks Optimization-Support Vector Regression (HHO-SVR). The study collects ozone concentration data as the response variable and meteorological conditions as predictor variables from a data website. The results show that seven of the 14 subdistricts produced the best predictions with an average Mean Absolute Error (MAE) less than 10, RMSE and MAPE less than 20, R^2 around 0.97, and Variance Regression (VR) around 0.98. The Diebold-Mariano test results show improved accuracy and performance stability of the HHO-SVR model in the subdistricts of Ciputat and South Bekasi, which are deemed well-suited for ozone prediction using the HHO-SVR model [55].

2.9 Principal Component Analysis (PCA)

PCA has shown promise in UHI assessment due to its ability to reduce the dimensionality of large datasets, while preserving the most important features of the original dataset. Before conducting a SWOT analysis on climate change response, a PCA was performed to reduce variables for distinct climate change responses to a limited number of components based on their common features. The Kaiser–Meyer–Olkin (KMO) and Bartlett tests evaluated the data and analytical technique, respectively. Using extracted variable factor scores, a cluster analysis was performed to determine the SWOT analysis factors. The PCA and k-medians non-hierarchical clustering were carried out using STATA 11.1. To evaluate the direction of urban expansion in relation to climate change, a non-hierarchical cluster analysis of k-medians non-layer clustering was conducted [56]. In another study, Landsat data from 2003 and 2017 was used to analyse LST (LST) in Accra, a sub-Saharan gateway to West Africa. Various ML

algorithms and spatial and statistical tools were utilized to analyse LULC and LST maps [57]. Madaus et al. aimed to anticipate an increase in the frequency and length of daily and multi-day heat waves. The UHI effect amplifies the consequences. By utilizing PCA, the authors found evidence for an increase in deep sea-breeze penetration in the New York City region by the end of the 21st century, which may help counter high heat episodes. The dataset may be analysed using ML techniques like PCA [58].

2.10 Kriging

According to Han et al., summer ozone (O_3) and UHI have negative effects on human health. To identify ozone and UHI distributions in Xi'an during the summer, the authors updated and modified a land use regression model using the RF ML technique. The study also employed a combination of land use regression and kriging to extract feature variables. Regression mapping was used to generate and spatially connect ozone and UHI distribution maps in Xi'an. The study used SHAP and PDP to explain the impact of different features on ozone and UHI [59].

2.11 Regression

Regression analysis serves as a valuable tool in investigating SUHI, LST, and UHI mitigation strategies. For instance, in SUHI studies, regression analysis enables modeling relationships between SUHI and environmental factors like impervious surfaces, population density, and green space. Terzi et al. employed supervised ML Ridge regression to analyze Istanbul's urban fabric's impact on SUHI generation, highlighting NDVI and building coverage ratio (BCR) as significant influencers, accounting for 71% of LST anomalies. Their findings underscore the pivotal role of urban features in SUHI dynamics, emphasizing the relevance of research and practice considerations [60].

Furthermore, regression analysis facilitates modeling relationships between LST and environmental factors such as land use/land cover, albedo, and vegetation. Duncan et al. utilized panel regression models to assess the cooling impact of different vegetation types across urban landscapes. Their study revealed that trees and shrubs exert a significant cooling effect, with a 1 km² increase in shrub (tree) cover leading to a surface temperature reduction of 12°C (5°C). Geographically weighted regression models demonstrated regional variations in the cooling impact, highlighting the influence of unobserved variables on temperature-vegetation relationships [61].

Moreover, regression techniques aid in understanding seasonal LST variations and their influential factors. Liu et al. identified metrics such as Mean Architecture Projection Area (MAPA), Building Coverage Ratio (BCR), Floor Area Ratio (FAR), and Mean Architecture Height (MAH) as key influencers of seasonal LST. Their study revealed consistent proportional contributions of these factors across seasons, elucidating their role in shaping seasonal temperature dynamics [62].

3 Discussion

The above-mentioned ML algorithms have been successfully applied to UHI assessment. Each algorithm is used in combination with a data source and a UHI related variable. Each combination possesses certain advantageous characteristics which should be suitably applied to the case. The UHI related variables covered in this study are shown in Figure 7.

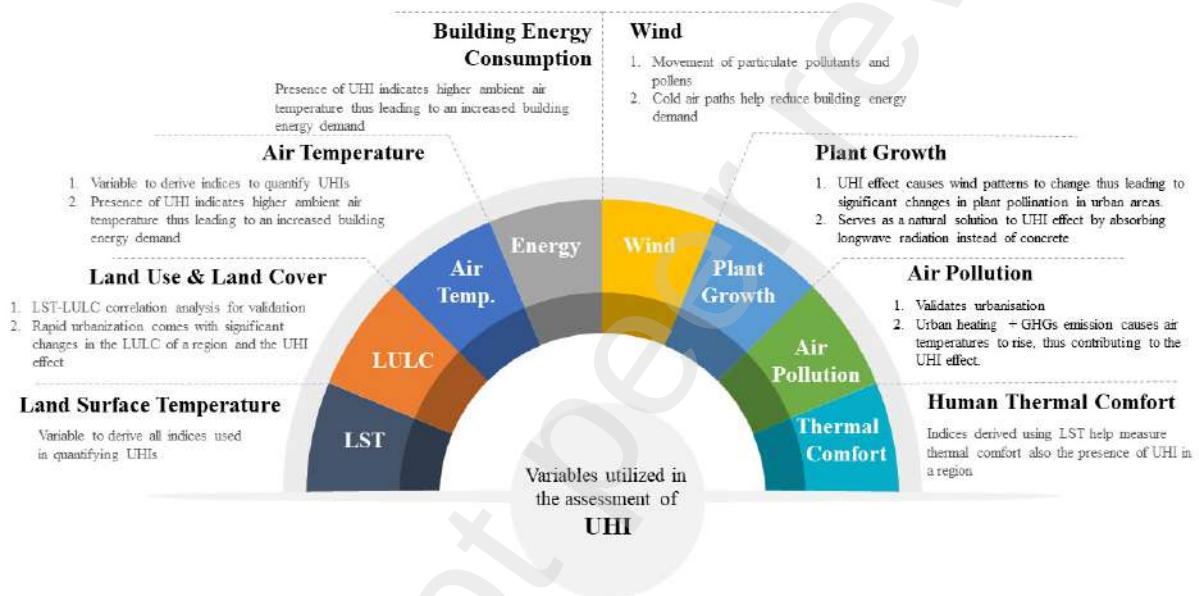


Figure 7 Variables utilised in the reviewed articles for UHI assessment.

To better understand the co-occurrence of UHI-related variables, data sources, and ML algorithms, we generated three 2D plots. The first plot illustrates the correlation between the UHI-related variable and the study data. The x-axis represents the UHI-related variable, while the y-axis represents the study data. Labels indicate the frequency of each unique pairing of the two variables, and a color scale denotes the prevalence of each combination (Figure 8). Two additional plots were created to investigate the relationship between the UHI-related variable and the ML algorithm used (Figure 10) and between the study data and the ML algorithm used

(Figure 9). Both plots use a similar labeling technique and color scale to denote frequency. The scatter plots facilitate qualitative and quantitative analysis, enabling the identification of patterns, clusters, and correlations. Taken together, these scatter plots provide a comprehensive visual representation of the relationships and patterns within the data, enhancing our understanding of the co-occurrence of UHI-related variables, data sources, and ML algorithms in existing research.

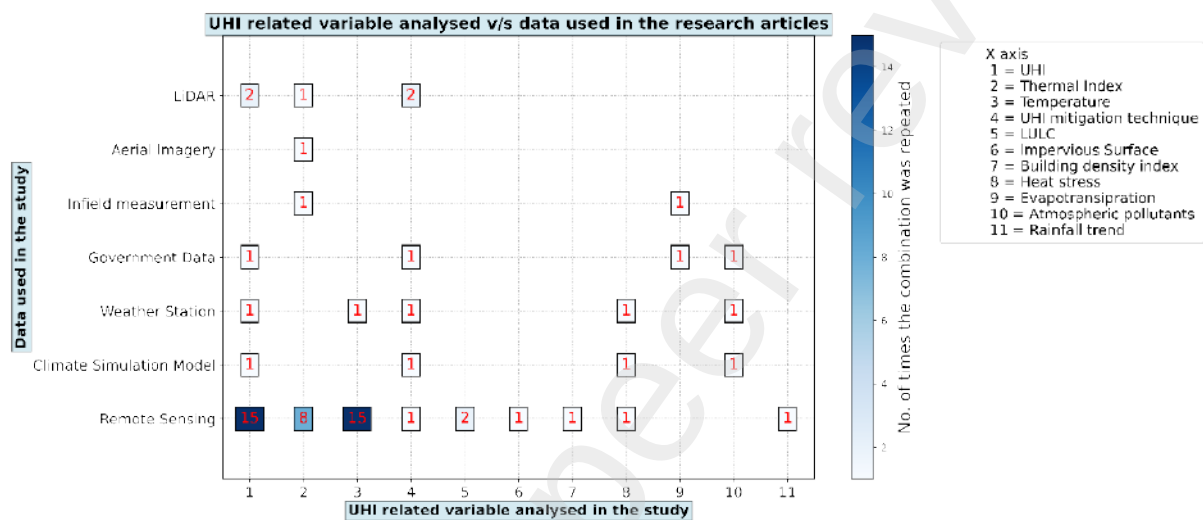
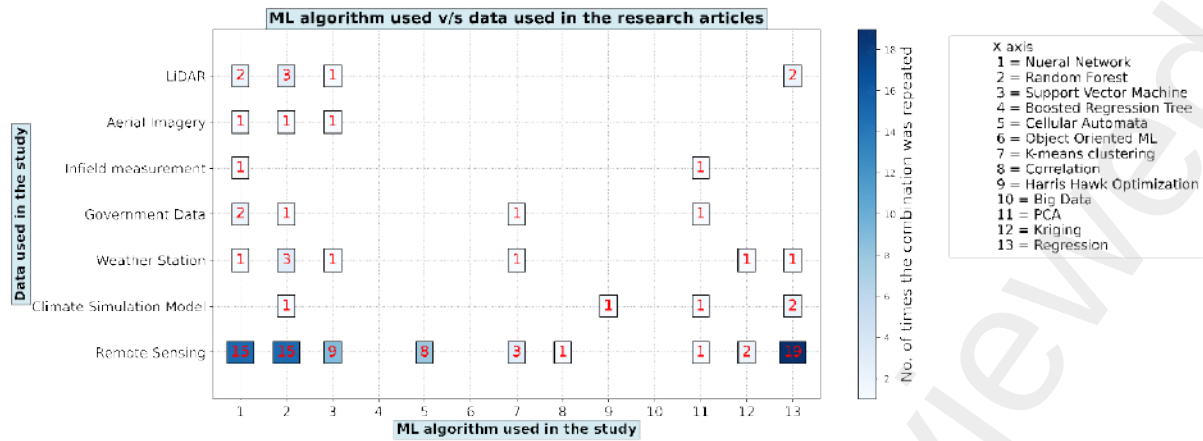
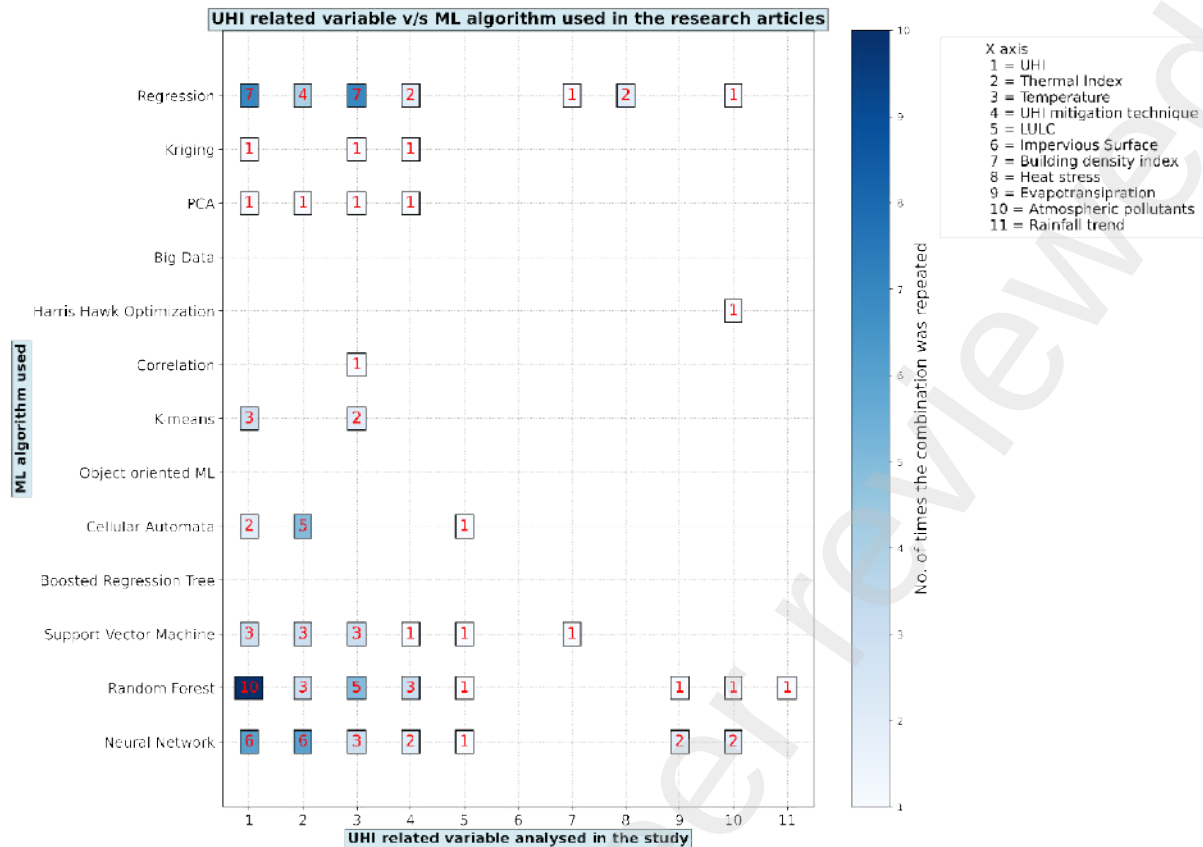


Figure 8 Two-dimensional scatter plot of the combination of UHI-related variables and data used in the analysis. The X-axis represents the UHI-related variables, ranging from 1 to 11 as indicated by the scale. The Y-axis represents the data used in the analysis. Each point on the plot represents a unique combination of UHI-related variables and data used in a research paper analysed in the study, with the number of occurrences of the combination labelled on the plot. The frequency of usage of different combinations highlights the patterns in the data and the interplay between UHI-related variables and data used in the analysis.



565

566 *Figure 9 Two-dimensional scatter plot of the combination of ML algorithm used and data used*
 567 *in the analysis. The X-axis represents the ML algorithm used, ranging from 1 to 13 as indicated*
 568 *by the scale. The Y-axis represents the data used in the analysis. Each point on the plot*
 569 *represents a unique combination of ML algorithm used and data used in a research paper*
 570 *analysed in the study, with the number of occurrences of the combination labelled on the plot.*
 571 *The frequency of usage of different combinations highlights the patterns in the data and the*
 572 *interplay between ML algorithm used and data used in the analysis.*



573

574 *Figure 10 Two-dimensional scatter plot of the combination of UHI related variable used and*
 575 *ML algorithm used in the analysis. The X-axis represents the UHI related variable used,*
 576 *ranging from 1 to 11 as indicated by the scale. The Y-axis represents the ML algorithm used*
 577 *in the analysis. Each point on the plot represents a unique combination of UHI related variable*
 578 *used and ML algorithm used in a research paper analysed in the study, with the number of*
 579 *occurrences of the combination labelled on the plot. The frequency of usage of different*
 580 *combinations highlights the patterns in the data and the interplay between UHI related*
 581 *variable used and ML algorithm used in the analysis.*

3.1 Understanding the Multidimensional Aspects of Urban Heat

Islands

This subsection discusses various variables in the context of UHI research. It starts with the commonly studied LST and LULC patterns, followed by remote sensing data limitations and the use of ELMs to improve LST calculations. It then covers energy balance concepts and the need to consider heat flow in 3D geometry. Finally, it delves into the importance of PET and the URG for UHI research.

The dominant focus of UHI studies has been LST and LULC patterns, often utilizing weather station and satellite air temperature data. However, such approaches neglect the impact of meteorological factors and urban geometries on air temperature at each measurement site, resulting in an incomplete assessment of UHI intensity and magnitude. To address these limitations, researchers have utilized ANNs to predict LST, LULC, and thermal metrics like the UTFVI for Ahmedabad in 2029 and 2039. The potential impacts of higher LST on city thermal comfort require further investigation. Although remote sensing data is commonly used to derive LST, its largest drawback is its resolution, which can be as large as 500m. To address this limitation, researchers can employ ELMs to lower the resolution of Landsat data from 60m to 30m, or use ML-based algorithms to increase the resolution of thermal images and achieve more precise LST calculations. Additionally, NDVI-based thermal sharpening and RF-based research can reduce ASTER imaging resolution from 90m to 10m by utilizing NDVI, NDBI, MNDWI, IS, and vegetation proportion. Ensemble learning and object-oriented ML techniques have also demonstrated the ability to enhance LST accuracy. Furthermore, architectural factors such as mean annual daylight availability, building coverage ratio, floor area ratio, and mean annual sunlight hours affect seasonal LST, as demonstrated by one study that used regression-based ML methods to examine their LST input. A regression tree-based study found that

606 elevation, NDVI, and LULC account for 86% of the spatial variability in LST within an area.
607 Geospatial terrain research with digital elevation models is essential for implementing
608 elevation-based mitigation measures.

609 Surface energy balance (SEB), terrestrial energy balance (TEB), and urban energy balance
610 (UEB) calculate heat fluxes to account for various heat sources and UHIs. The study found that
611 these concepts are underutilised compared to conventional UHI analysis methods, but they
612 provide a more complete view of UHI occurrences.

613 Heat flow and UHI categorization often ignore a city's three-dimensional geometry. To
614 quantify heat source features that impact UHI intensity and magnitude, a multidimensional
615 analysis of the city where UHIs occur is needed. A study proposed ten architectural metrics,
616 based on ten BRTs, to characterise 3D building patterns in a Shanghai neighbourhood, while
617 another innovative method helps urban planners identify 2D and 3D elements that affect urban
618 surface temperature. Future study may interpret results based on urban form volume. It's
619 important to rethink this strategy and create a new UHI model that accounts for cities' complex
620 three-dimensional shapes and heat emissions, which affect UHI size and intensity.

621 Using the PET gauge, air temperature, and urban humidity, ANNs can study heat perception.
622 Another study examined urban design and PET score for optimal thermal comfort. ANN-based
623 UHI assessments also consider thermal comfort-related environmental factors like relative
624 humidity, wind direction and speed, cloudy sky index, diffuse horizontal irradiance, and global
625 horizontal irradiance. ANNs can also analyse internal warmth and air pollution, which affect
626 heat gain.

627 The URG is essential for assessing UHIs. Urban core temperatures are much higher in UHI
628 zones. ML can compute this gradient. One study used surface emissivity maps from NDVI-

based Landsat data for RF-based SUHI effect and pattern analysis. The study found that urban air temps make URG monitoring easier because concrete and asphalt slow night-time cooling.

3.2 Addressing Urban Heat Island Effect: Strategies and Mitigation Measures

In this subsection, we delve into various aspects of UHI research. Firstly, we discuss UHI intensity measurement, followed by predicting the urban scenario, and then the impact of urban vegetation. Next, we explore plant propagation and policy-making for mitigation. Lastly, we discuss green infrastructure-based mitigation, along with other temporary mitigation measures. Through this order, we provide a comprehensive overview of UHI research and the various measures that can be taken to mitigate its impact on urban areas.

UHI analysis and mitigation require intensity measurement. Boltzmann identifiers allow this. Remote sensors, flux tower, black box, and white box ML can simulate UHI. Building height and albedo increase SUHI size and intensity. CA-based neural networks can study LST, LULC, and UTFVI. QGIS' six-step projection method modelled a future growth scenario, which could aid mitigation efforts. Urban microscale temperature forecasts and K-means algorithm-based temperature attenuation curves reveal city growth trends and UHI intensity. K-means, LST, and city-specific LCZ clustering can help academics find regional UHI patterns. K-means clustering and CDI-based studies show that urbanisation raises temperature, laying the groundwork for a sustainable urban plan to combat UHI. A big data-based urban ecological grading system was examined. Kriging tracks UHI. UHI and ozone spread health impacts were studied in summer. Finally, SUHI-based regression-based ML helped find UHI trends.

Predicting the urban setting is crucial for planning. In this paper, ML algorithms like ANNs, RNNs, DNNs, RF, SVMs, BRTs, and CA predict variables like LST, LULC, UTFVI, building density, etc. Local factors and training dataset availability determine the method. PCA forecasts

653 daily and multi-day heatwaves. This can predict extreme heat events and lower high-
654 temperature deaths. Some studies use RF and BRT to forecast ($T_{\max}$) in a place to estimate
655 mortality rate and lessen such tragic fatalities. Large-scale ML prediction research will help
656 analyse future scenarios and prepare UHI impact countermeasures.

657 Urban vegetation reduces the UHI impact and improves stormwater control, pollutant
658 concentrations, and biodiversity. ML-based algorithm studies of green infrastructure dynamics
659 help maintain and study urban green spaces. In a regression analysis study, trees and bushes
660 outperformed grass in cooling infrastructure systems. Another study used RF to examine the
661 cooling ability of urban green areas based on tree species, height, LAD, crown base, LAI, trunk
662 circumference, street direction, and crown area. Another study examined how deciduous and
663 coniferous trees affect urban building cooling loads. The RF algorithm helped build sustainable
664 urban areas. SVMs also spot urban tree species.

665 UHI also affects plant propagation. ML algorithms using LST data and phenological
666 observations can assess this effect. Regression-based algorithms can predict a city's plant
667 species' blooming start date. Interestingly, the implementation of green roofs, which utilize
668 vegetation to mitigate UHI, represents a potential solution to this phenomenon. However,
669 recent research suggests that UHI may in fact be impeding the effectiveness of green roofs as
670 a UHI mitigation strategy.

671 Finally, UHI mitigation needs stakeholder cooperation. This article shows that ML helps
672 evaluate UHIs and execute optimal mitigation strategies. Researchers must use ML-based UHI
673 analyses to help governments and policymakers mitigate UHI effects. To counteract the
674 negative impacts of the UHI effect, several potential solutions have been proposed for
675 stakeholders to implement. One potential solution is the development of a nationwide UHI
676 directory based on clustering using UHI-related variables such as LST, SUHI, air temperature,

and LULC change. This directory would facilitate targeted UHI mitigation efforts and help prioritize areas most vulnerable to UHI effects.

A potential solution is the implementation of Green Infrastructure, following the example of Canada's low impact development policies. This could involve the gradual transition towards compulsory implementation of low-impact development techniques, such as green roofs and permeable pavements, as a way to increase urban vegetation and reduce the heat island effect.

Temporary mitigation measures, such as the use of reflective paints and shading, could also be implemented to reduce the absorption and re-radiation of heat in urban areas. Regular monitoring and analysis of UHI could further aid in the identification of risk zones and the prioritization of mitigation efforts.

Additionally, studies could be conducted to identify plant species, substrates, and irrigation methods best suited for extracting thermal benefits from green roofs in tropical environments such as India. Moreover, studies could also be conducted to understand the effect of green roofs on building energy consumption, as green roofs can have both cooling and insulating effects. Overall, implementing these potential solutions could aid in mitigating the negative impacts of UHI and promoting sustainable urban development.

3.3 Advancing Urban Heat Island Research through Machine Learning and Three-Dimensional Modeling

ML has the potential to improve the accuracy and scope of UHI assessments, including the prediction of future occurrences of UHI indicators such as LULC and LST [28]. Nevertheless, this review identifies gaps in UHI research, such as disparities in the spatial range of meteorological data and UHI classification criteria. To overcome these challenges, future studies should prioritize matching the spatial range of meteorological data with the UHI

classification criteria, consider equipment measurement installation criteria, and analyze three-dimensional urban geometry factors. Three-dimensional modeling and overlay analysis can aid in achieving these goals. In particular, predictive modeling could allow for scenario-based UHI analysis using ML in 2030, 2050, and 2070, which are essential in India's objective to achieve net-zero emissions. The use of remote sensing, field weather stations, and mobile traverse for data collection should also be considered. Furthermore, future studies should account for additional horizontal and vertical three-dimensional urban geometry factors in UHI areas. Overall, this review emphasizes ML's potential in UHI assessment and highlights the need for further research to enhance the accuracy and scope of UHI analysis.

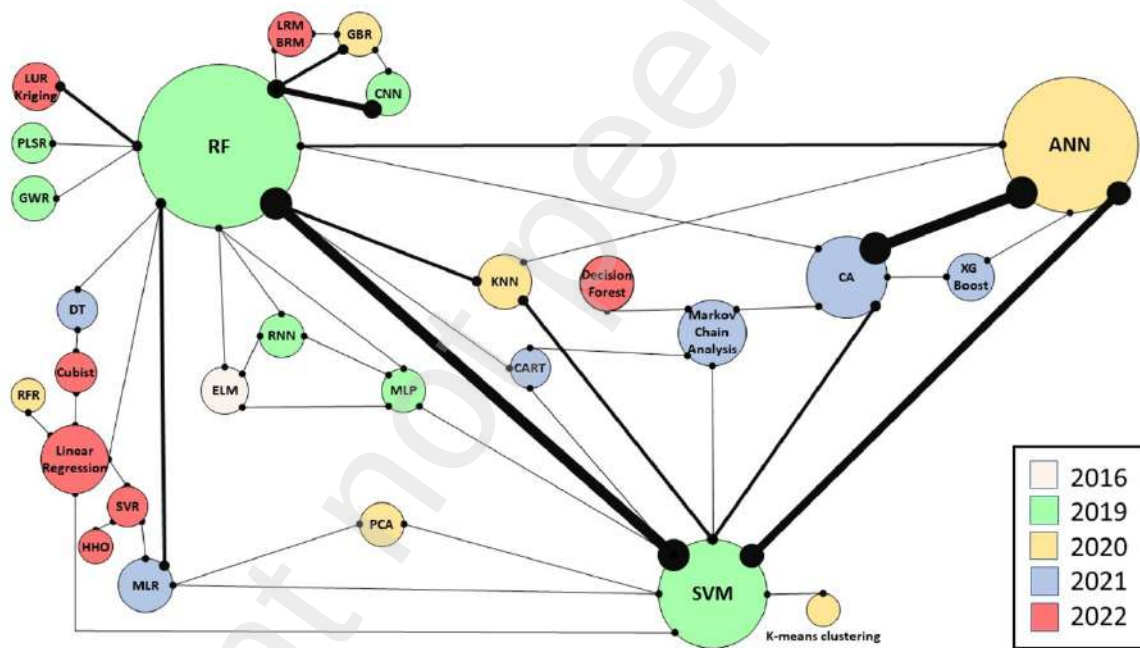


Figure 11 Frequency of use of ML algorithms in the analysed publications. The size of the circles represents the relative frequency of use of each algorithm, with larger circles indicating greater use. Connecting lines between circles indicate the use of multiple algorithms together, with line thickness representing the frequency of use. The color of the circles represents the first year of use for each algorithm within the selected publications, as indicated in the legend.

715 *This figure provides a qualitative analysis of the utilization of ML algorithms in the reviewed*
 716 *articles related to UHI assessment.*

717 **3.4 Visualization and Analysis of Machine Learning Algorithms** 718 **in Urban Heat Island Assessment: Insights and Challenges**

719 In this study, Figure 11 visualizes the frequency of ML algorithm use in analyzed publications
 720 related to UHI assessment. The figure utilizes circles to represent each algorithm's relative
 721 frequency of use, with larger circles indicating greater use, and connecting lines between circles
 722 indicating the use of multiple algorithms together. Circle color indicates each algorithm's first
 723 year of use, as shown in the legend. This qualitative analysis provides insight into the utilization
 724 and collaboration of machine learning algorithms in UHI assessment studies. The figure
 725 suggests a need to adopt more advanced and sophisticated techniques as recent algorithms offer
 726 improved performance and accuracy. Additionally, combining techniques for more
 727 comprehensive analysis is a growing trend. While ML proves effective in UHI assessment,
 728 challenges still exist. The following section discusses these challenges.

729 **3.5 Challenges**

730 While ML holds promise for advancing UHI assessments, several formidable challenges
 731 impede its effective deployment within this domain. Firstly, the acquisition and curation of
 732 high-quality, voluminous data, encompassing variables such as temperature, humidity, and
 733 land use, represent significant hurdles. The veracity and completeness of these datasets are
 734 paramount for robust model training and validation. Moreover, the absence of standardized
 735 protocols for data collection and processing hampers comparability across studies and regions,
 736 thwarting efforts towards synthesis and generalization.

737

Secondly, the intrinsic complexity of the UHI phenomenon, influenced by multifaceted factors like urban morphology, anthropogenic heat generation, and microclimate dynamics, poses a formidable modeling challenge. ML models must grapple with this intricate interplay while striving for predictive accuracy and interpretability. Thirdly, the interpretability of ML models, crucial for discerning the causal mechanisms underlying UHI dynamics, remains a pressing concern. Enhancing model transparency and elucidating feature importance are imperative for fostering stakeholder trust and facilitating informed decision-making.

Moreover, the paucity of ground truth data for UHI validation impedes model assessment and calibration, exacerbating uncertainties in model outputs. Additionally, ML models' limited generalizability across diverse spatiotemporal contexts and changing environmental conditions underscores the need for adaptive, transferable modeling frameworks. Furthermore, the interdisciplinary nature of UHI research necessitates concerted efforts to bridge domain-specific knowledge gaps and foster cross-disciplinary collaborations.

Notwithstanding these challenges, the application of ML in UHI assessments harbors immense potential for enhancing urban resilience and sustainability. ML algorithms offer unparalleled capabilities for elucidating complex spatiotemporal patterns, guiding evidence-based policy interventions, and informing urban planning strategies aimed at ameliorating UHI's adverse impacts. In the ensuing subsection, we delve into the transformative opportunities afforded by ML in UHI research, delineating innovative methodologies and promising avenues for future inquiry in this critical domain..

3.6 Opportunities

The utilization of ML methodologies in UHI assessment heralds a profound opportunity to augment our comprehension of this intricate phenomenon and to devise more efficacious

strategies for its amelioration. The potential avenues for leveraging ML in UHI assessment are manifold and encompass several key domains:

1. High-resolution mapping: ML algorithms offer the capability to generate high-resolution maps of UHI, furnishing urban planners and policymakers with indispensable insights into spatial UHI distributions and facilitating targeted intervention strategies.
2. Predictive modelling: ML techniques enable the development of predictive models to anticipate future UHI scenarios, equipping decision-makers with foresight to preemptively address impending UHI exacerbations.
3. Identifying key factors: ML can discern the salient factors underpinning UHI manifestation within specific locales, furnishing policymakers with actionable intelligence to formulate targeted mitigation policies.
4. Integration with diverse data sources: ML can seamlessly integrate diverse data streams, encompassing remote sensing, meteorological, and socio-economic data, engendering a holistic understanding of UHI dynamics.
5. Automation of intricate processes: ML-driven automation expedites complex tasks such as image classification and data analysis, fostering efficiency gains and minimizing human fallibility.
6. Real-time monitoring: ML enables real-time UHI monitoring, facilitating prompt detection of emergent UHI intensifications and enabling timely intervention.
7. Cost-effectiveness: ML-driven model development engenders cost-effective solutions, characterized by scalability and applicability across diverse geographical contexts.
8. Personalized recommendations: ML-driven analytics can furnish personalized UHI mitigation recommendations tailored to specific locales, optimizing intervention efficacy.

The convergence of burgeoning data availability and ML algorithmic advancements portends continued strides in harnessing ML for UHI assessment, auguring transformative contributions to urban sustainability and resilience.

4 Conclusion

This study presents a comprehensive review of machine learning approaches for UHI assessment. These approaches have been widely used in other fields like image classification, speech recognition, natural language processing, and have shown promising results in UHI assessment. These models can effectively evaluate UHIs by predicting UHI indicators such as land surface temperature, air temperature, and LULC. They can also be used to predict future occurrences of UHI indicators and help overcome the limitations of traditional meteorological data collection methods for UHI computation and analysis.

We have analyzed and presented in detail the advantages and disadvantages of various machine learning approaches for UHI assessment. Special emphasis is given to hybrid models, which combine two or more machine learning techniques in a way that each model complements the strengths of the other. Various combinations of hybrid models and their novelty are identified in the literature and systematically presented in this paper. We have also identified gaps in current UHI research, such as mismatches in the spatial range of meteorological data and the UHI classification criteria.

The use of machine learning models for UHI assessment is expected to grow in the future, in line with the global drive towards sustainability. This paper provides a comprehensive summary of the existing machine learning approaches for UHI assessment and paves the way for future research in this field. We recommend further studies to address the identified gaps, prioritize matching the spatial range of meteorological data with the UHI classification criteria,

consider the installation criteria for measurement equipment, and include three-dimensional urban geometry factors in their analysis. These studies will help to improve the accuracy and scope of UHI assessment, which is critical for effective urban planning and management.

4.1 Cost implications and future scope

The adoption of machine learning techniques for UHI assessment requires extensive data collection, monitoring, modeling, and control networks. These infrastructure requirements inevitably add to the overall operating costs. The challenge here is to understand the added benefits in terms of the cost of investment and the cost regained due to energy savings arising from the adoption of UHI mitigation strategies. However, limited studies have investigated the financial implications of UHI mitigation techniques due to the confidential nature of data involved or market forces.

Some studies have highlighted the costs associated with UHI mitigation, such as the deployment of sensors and the implementation of control strategies. For instance, a study by Santamouris et al. showed that cool roofs and green roofs could potentially reduce energy consumption in buildings by 6-20%, depending on the climate zone. The cost of installing such systems, however, may not be feasible in some regions. Another study by Li et al. evaluated the cost-effectiveness of green roofs and found that the return on investment varied depending on the size of the building and the geographical location.

While some studies have focused on the financial implications of UHI mitigation, many challenges still exist in the implementation of machine learning techniques for UHI assessment. For example, the cost of deploying sensors and developing UHI models may be prohibitive, especially in developing countries. Additionally, the lack of awareness and understanding of the benefits of UHI mitigation techniques can hinder their adoption. Furthermore, there is a need for standardized methods and protocols for data collection, analysis, and reporting.

Despite these challenges, the use of machine learning techniques for UHI assessment has great potential for improving urban sustainability and mitigating the adverse effects of climate change. Future research should focus on developing cost-effective and scalable UHI mitigation strategies that can be implemented in various urban contexts. Additionally, there is a need for interdisciplinary research collaborations to better understand the social, economic, and environmental impacts of UHI mitigation strategies.

5 Declaration of Generative AI

During the preparation of this work the author(s) used ChatGPT in order to improve the readability and English language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

6 Table

Table 1 Overview of reviewed studies on the application of ML algorithms in the estimation of UHI-related variables. The table includes columns detailing the ML algorithm used, the UHI-related variable studied, the data used, the results obtained, the limitations of the methodology, and the merits of the methodology, as well as the reference of each study.

ML Algorithm	UHI related variabl e	Data sources	Results	Merits of the methodology	Refer ence
ANN	PET	Infield measure ment	optimal tourist opportunities in Mt. Ainos, Greece, from May to September, with highest frequency of thermal comfort appearing over the highest altitudes.	PET index was estimated using ANNs in ten sites within the Ainos National Park.	[16]
SVM, CA, ANN	SUHI, LST	Landsat 5, 7 & 8	predicted a substantial increase in urban built-up areas (9.23% in 2029, 13.59% in 2039), with higher temperatures in built-	CA model has excellent accuracy with an overall kappa value of 0.82, and the ANN	[13]

			up areas and a strong correlation between changes in urban areas and rising LST.	model produces excellent results with low error (MSE) and high correlation coefficient.	
CA, ANN, XGB regression model	UTFVI, predictive analysis	Landsat 5 & 8	predicts a 5.77% and 13.08% increase in built-up area, a 4.15% and 12.54% reduction in cropland area, higher LST ranges greater than 45°C in summer and 35°C in winter in 2025 and 2030, and strongest UTFVI zone in surrounding rural areas.	Accurately predicts land use, LST, and UTFVI, enabling urban planners to formulate effective UHI mitigation strategies.	[11]
CA, ANN	UTFVI, predictive analysis	Landsat 4, 5 & 8	predicts 8-11% urban area increase and 9-14% green cover decrease in Cumilla City by 2029 and 2039. Average temperature rises of 4°C and strong UTFVI effect forecasted.	Combines LULC with UTFVI derived from LST and validates the prediction for 2029 & 2039 by using ANN.	[12]
KNN, ANN, SVM, RF	LST IS, URG	MODIS, Landsat	significant increase in IS area 2000-2019, strongly correlated with mean LST. Urban-rural mean LST showed average difference of -3.59°C during day and 2.33°C at night, indicating a cool island effect during day and UHI effect at night, both growing stronger from 2000-2019.	The study found correlations between LST, impervious surfaces, green spaces, and bare lands in Greater Cairo to mitigate increasing SUHI through urban planning.	[63]

CA, ANN, SVM	UTFVI, LST, LULC	Landsat 5 & 8	the conversion of 27.24% bare land and 5.43% vegetation into built-up areas in Kuwait from 1991 to 2021 led to a 5°C increase in mean LST and a 0.861 increase in UHI, with the strongest UTFVI effects seen in 108% of the area	2031 predictions using ANN-based CA show decline in vegetation, water bodies, increase in built-up areas, and higher LST with 41% in UTFVI affected zones validated with high accuracy indicated by RMSE and R ² values.	[15]
SVM, ANN, CA	LST, UTFVI, LULC	Landsat 5, 7 & 8	Rapid urbanization raised mean LST by 5°C, UHI by 0.861 and UTFVI by 108% from 1991-2021 & decline in vegetation and water bodies by 2031 with 41% of the area having strong UTFVI effects, validated by RMSE and R ² values.	Comprehensive analysis predicts LULC, LST, UHI, and UTFVI changes in Kuwait 1991-2031 with high accuracy validated by RMSE, R ² values.	[15]
ANN	E-UHI	Landsat 8, WorldView-3	Study finds the scale, shape and type of factories impact E-UHI with higher LST in high-temp. factories. The Δ LST of green space, staff activity, material transfer and storage, factory building, smelting, and casting building are 3.95-7.32°C.	ML simulates and quantifies impact of factory components on E-UHI, providing comprehensive analysis and optimal ranges for mitigation.	[18]

RF, CNN	UET	Berlin Digital Atlas	producing results with a RMSE of 0.0239 mm/h & R^2 of 0.840 (RF) & RMSE of 0.0250 mm/h & R^2 of 0.824 (1DCNN) for a more vegetated site, demonstrating its potential in response to climate change.	Integrating meteorological and GIS data enhances accuracy of ET predictions, promising for modelling in heterogeneous urban environments.	[20]
BPNN	UBCE	Governm ent dataset	predicts UBCE based on BEF in Changxing, with MAPE of 10-20%. High UBCE clustered in middle-east and south of city centre, with carbon emission map produced	Study showed impact of built environment factors on urban BEC and hotspot clustering in urban planning.	[21]
ELM	LST	MODIS, Landsat	achieving smaller average difference (1.7 K), high correlation coefficient (0.755) and low root-mean-square error (1.824).	Combines MODIS and Landsat product using ELM algorithm to produce high resolution LST dataset aiding in analysing the UHI effect.	[22]
Boltzman n Machine	SUHI, UHI distribution	Landsat 5, 7 & 8 weather data	The improved RBM identification algorithm had a high accuracy of 93.31% and better clustering effect compared to K-means and genetic K-means algorithms.	builds a heat island intensity identifier using improved RBM with weak supervision, providing accurate support for urban planning and security. Test time was 0.72s	[24]

DNN, U-NET	UHI mitigation	Aerial Dataset	if all the selected rooftops installed green roofs, 1.210465×10 ⁶ kg/year pollutants would be cleaned up with dry deposition.	94.17% validation accuracy	[25]
RF, SVM, MCA	LST	Landsat 7 & 8, ASTER DEM	Accuracy was more than 94.5% for all but barren land class.	Three ML methods were compared and the one with highest accuracy was used for predictive analysis.	[29]
RF	URG	Landsat 5,7 & 8	Increase in urban cover from 341.72 km ² to 520.58 km ² and average LST increase from 17°C to 38°C. Negative relationship between NDVI and LST (-0.95) found.	Studies the effect of UHI on Ibadan city using a combined LULC change and URG approach.	[64]
RF	Rainfall trend	Sentinel 2A	Correlation between rainfall trends and UHI effect, with increased rainfall associated with UHI and decreased rainfall due to low humidity and aerosols.	Correlated UHI phenomenon with urban flooding	[65]
RF	SUHI, URG	Landsat 8	Kathmandu's UHI effect is remarkable and correlated with the landscape composition and pattern, with lower green space density in the city core.	Studies the effect of SUHI on a valley in Kathmandu city using a combined LULC change and URG approach.	[30]

LR, SVM, RFR, SVR	buildin g density index	Landsat 5, 7 & 8	Regression performed best with a large sample size and LR was better for limited sample size with a positive linear relationship between predictor and building density.	determines the best model for Yogyakarta's built-up expansion and densification over multiple decades.	[66]
RF, SVR, MLR	SUHI	MODIS, LIDAR	high accuracy predictions ($R^2=0.96$ and $RMSE=0.83$) through RFR, with building height and albedo identified as the main drivers of heat trapping. Retrofitting with green infrastructure is proposed.	Combines explanatory variables like Skyview factor, elevation, urban form etc. with response variable LST to map the SUHI using 3 ML methods	[35]
LUR-Kriging-RF	Spatial distribu tion of UHI	Landsat 8 & Meteorol ogical station	Higher R^2 (0.65 for ozone, 0.93 for temp) & lower RMSE (28.18 for ozone, 0.92 for temp) compared to traditional LUR (0.4 R^2 for ozone, 35.66 RMSE, 0.8 R^2 for temp, 1.52 RMSE).	model outperforms traditional LUR in mapping ozone pollution & urban heat, resulting in improved R^2 & lower RMSE.	[59]
RF	UTCI, human	In field meteorol ogical	Thermal sensation & comfort in urban park affected by RH & Ta in winter & Ta in summer, varies by landscape type, with neutral temperatures 12.83°C & 18.18°C in winter and	ML used to examine human thermal comfort in an urban park, determining thermal	[37]

	thermal comfort	measure ment	15.12°C & 24.75°C in summer, temperature acceptability range 2.8-21.9°C (PET) & 8.1-28.3°C (UTCI) at 80% acceptability.	benchmarks for sustainable design and providing reference.	
RFR	UTFVI, UI, IBI etc.	Landsat 8, LiDAR	develops regression models to estimate UHI intensity with 89% variance explained and a root MSE of 0.23°C.	Uses 33 explanatory variables and provides high accuracy.	[67]
RFATPK	LST	Sentinel 2, ASTER	geostatistical downscaling method outperforms thermal sharpening method in accuracy and detail, when downscaling 90m ASTER LST to 10m by considering nonlinear LST & independent variables, and regression residual LST scale effect for improved LST.	The RFATPK method produced more accurate LST data compared to the TsHARP method, specifically capturing LST differences in heterogeneous regions with improved spatial details.	[36]
RF	LST, UHI mitigation on	In field measure ment	The trunk circumference of trees has been found to be a valuable indicator for estimating the climate regulating ecosystem services.	ML analysis identifies important tree physiognomy traits for mitigating UHI effect, with results strengthened by diverse climate zones and multiple measurements.	[34]

SVM, RF, MLP	UHI mitigation on	LiDAR, Ground truth	resulting in overall accuracy of 80% for SVM, 83.75% for RF, and 73.75% for MLP.	The SVM and RF algorithms performed better than MLP in identifying urban tree species.	[40]
MLR, RF	UHII	Sentinel 2, Weather stations	RF models predict UHII with 96% accuracy, main predictors being land cover and meteorological variables.	Research provides UHII maps, identifies contributors, and suggests mitigation based on green structures and asphalt's impact on air temperature.	[39]
RF, GWR	UHI, LST, mitigation on	MODIS	an increase in shrub cover reducing surface temperatures by 12°C and an increase in tree cover reducing temperatures by 5°C.	Fine-res imagery and temperature data analysed for urban vegetation relationship using statistical models, including regression, GWR, and ML, trained with city-wide data.	[61]
RF, CNN, GBR	LST	Landsat	Increasing vegetation and reducing impervious areas can reduce LST by up to 10°C. Advanced statistical models were used and accurate LST prediction is possible.	The research uses advanced nonlinear statistical models and geostatistical data to analyse the interdependent effects of urban characteristics on urban surface temperature.	[32]

GBR, RF	LST, Air tempera ture	Meteorol ogical station, MODIS, ASTER	model with 5 predictors (LST, Julian day, NDVI, DEM, SZA) was the best performing regression model in predicting $T_{\max}$ in London at a high spatio-temporal resolution, with a R^2 of 0.68, MAE of 1.60°C, and RMSE of 2.03°C.	The methodology provides a framework to produce high spatio-temporal resolution $T_{\max}$, aiding public health researchers and informing decision-making towards mitigating heat-related mortality.	[33]
RF, SVM, PLSR	LST	MODIS	The response rate to temperature showed a slope of 0.708 within the upper asymptote and 2.148 in the middle range, with the lower asymptote likely related to extreme warm spells.	The study used multiple regression models for prediction with the best being RF, with a high R^2 (0.829) & low error (1.461 days).	[68]
RF, ANNs, SVM	SUHI, LST, LULC, UTFVI	Landsat & UAV	The MLP model outperforms other LC classification algorithms with very good accuracy, while the SVM classification produces almost the same results with slightly lower accuracy.	compared satellite-derived and UAV-derived LST	[31]
ELM, RNN, MLP, RF	Ozone	Weather station	RNN outperforms other ML models and traditional atmospheric models in simulating tropospheric ozone, achieving high accuracy and low cost	Study predicted tropospheric ozone in using WRF-CMAQ, ML models, meteorology, air	[69]

				pollutants more accurately. Dew-point deficit, NO2 main factors,	
SVM	LST, UTFVI, LULC	Landsat 4, 5, 7, & 8	highest recorded temperatures in Ahvaz occurred in areas of bare land and residential development, 42.93°C & 40.06°C respectively.	Combines the LST and LULC to perform a change detection for 2016 for both LULC and UTFVI which is derived from LST	[47]
Supervised learning MLC	UHA	Landsat 4, 5, 7, 8	Observed highest LST in built-up class. Overall accuracy of 92%, 89%, and 91% achieved for three different years.	Combines the LST and LULC analysis to identify urban hotspots. LST analysis is done for each of the different LULC classes separately.	[70]
LRM-BRM, RF, GBR	UTEI	MODIS	better urban form improves the quality of urban environments, with an average decrease in UTEI of 0.69°C in summer and 0.56°C in winter	Considers the contribution of urban form factors as well as natural factors to the heat island effect.	[71]
Boosted Regression	LST, UHI metrics	Landsat 5, 7 & 8	predicts LST with a correlation of 0.956. Optimal urban cooling involves a balance of 60% natural land in 7-8 patches per km ² , resulting in a 2.1°C reduction in building surface temperature.	emphasizes importance of both landscape composition and configuration in urban cooling planning to accommodate population growth and climate change.	[72]

Boosted Regression	UHI mitigation	ECLMS	predict the spatial distribution of cold-air paths, resulting in promising results with AUC values between 0.85 and 0.96, and three predictor variables related to surface elevation having strong influence on cold-air path occurrence.	Study presents an efficient method to identify cold-air paths covering large areas, identifying important predictor variables for occurrence without time-consuming field measurements.	[44]
Boosted Regression	LST	Landsat 8	3D urban architecture impacts LST in Shanghai, with key factors being BCR, MAH, height deviation, and projection area, regulation amplitudes of 2.7°C, 0.9°C, 0.6°C, 0.6°C respectively.	Uses 10 architectural metrics to describe 3D building patterns, providing quantitative insights for mitigating UHI via rational design and management.	[45]
Boosted Regression	LST, LULC	Landsat 8, Worldview 3	both 2D and 3D indicators play a role in shaping the surface temperature and 3D indicators have a more significant impact in different urban geometries.		[46]
Cubist, GBDT, SLR	AHE	MODIS, Government surveys	model proposed using Cubist algorithm with prediction error $<0.9 \text{ W/m}^2$, effectively captures spatiotemporal building heat variations and accurately estimates AHF in urban regions.	model uses multi-source data, small samples for AHF representation, verified to effectively capture building heat variation & reduce error when compared to regression.	[73]

CA, ANN	LST, UTFVI, LULC	Landsat 5, 7 & 8	predicts a 42% to 44% increase in built-up areas, causing 13-14 km ² stronger UTFVI zones by 2025-2030, and decrease in green cover, bare land, and water bodies by 21%, 20%, and 1%.	The study provides reliable predictions for 2025 and 2030 and effective strategies for mitigating UHI effects.	[48]
CA, ANN	LST, URI, SUHI	Landsat 5, 7 & 8	predicts a 34.78% reduction in vegetation cover and an increase in built-up areas, leading to a rise in high and very high UHI-affected areas by 25.63% and 7.58% in summer 2030 and moderate UHI effects in 55.33% of the area in winter 2030.	Landsat data and correlation analysis enhance study's findings and provide useful tool for urban planners.	[14]
K-means clustering	SUHI	MODIS LST	analysed the proportion of 6 land-use types in 3 levels of cities and found built-up areas and farmland to be the main components, with built-up land increasing and farmland declining yearly.	Combines land use, population and LST to do an overlay analysis and clustering analysis of temperature attenuation curve.	[74]
K-means clustering, SVM	UMTF	ASTER, Weather station	achieving an average improvement of 14.8% and 10.4% in $T_{\max}$ & $T_{\min}$ forecasts in summer and reducing the forecast	UMTF reduces mean absolute errors (1.0°C) & improves correlation ($R^2 = 0.84$) showing	[50]

			bias, with a reduction of 0.3°C to 1.03°C in MAE compared to traditional models.	efficiency in providing high-res temp forecasts in heterogeneous urban environments.	
K-means clustering	UHII, SUHII	ECLMS	GIS-based LCZ mapping outperforms remote-sensing-based WUDAPT & identifies complex city-specific LCZ transitions.	GIS-based LCZ mapping improved with fuzzy logic and building height for increased accuracy over WUDAPT L0.	[75]
K-means clustering	SUHI, LST	Landsat 5, 7 & 8	Sustainable development policies affected UHI patterns and intensities, leading to less aggravated UHIs in the "recent" new towns	Compares UHI dynamics using climate and demographic variables for 4 towns at a time giving a holistic approach to UHI mitigation	[49]
K-means clustering	SUHI, LST, CDI	Landsat 5 & 8	The results of the study show a correlation between the increase in temperature and built environment changes, and the use of CDI for spatialized assessment of urban landscape changes in terms of surface temperature increase.	provides a low-cost tool for integrated and spatialized assessment of urban landscape changes through the use of remote sensing images. The Change Detection Index (CDI) summarizes NDBI and LST changes.	[52]
	UHI zone	Landsat 8	Direct correlation between OTC and UHI zonings. Recommended a comprehensive formula for summer and winter OTC.	Urban form zoning based on Outdoor Thermal Comfort; Method can be applied to other cities	[53]

Statistical correlation, MLC	UHA	Landsat 8	Positive correlation between LST and urbanization ($R^2=0.61$) and industrial development ($R^2=0.27$), and a negative correlation between LST and vegetation ($R^2=0.20$ and 0.15 for NDVI and SAVI respectively).	Combines the LST based hotspot analysis with an index based statistical correlation for better validation of results	[54]
SVR, HHO		weather. com	Model had MAE <10, RMSE, MAPE <20, R^2 around 0.97 & VR around 0.98. Model was more accurate & stable for Ciputat & South Bekasi sub-districts.	combines HHO with SVR to improve results	[55]
PCA	UHI mitigation on	Meteorological data	Provides park indicators as guidelines to install green infrastructure in response to climate change.	Combines clustering, SWOT & PCA to give a holistic approach	[56]
RF, KNN, SVM	SUHI, URG	Landsat 7 & 8	The urban-rural LST difference was 1.4°C in 2003 and 0.28°C in 2017, with a strong correlation between land use/cover and mean LST, considering factors such as mean size, shape, largest patch, and aggregation.	combines ML techniques, spatial and statistical methods to analyse the impact of LULC on LST in Accra, giving valuable insights for urban planners	[57]

PCA	UHI distribu tion	GCMs	Results validate system's scalability & efficiency, providing downscaled predictions for infrastructure resilience and emergency response.	Hybrid downscaling combines stat & ML for efficient, detailed, scalable results for heat stress in cloud architecture.	[58]
Ridge regression model	SUHI	Landsat 8	71% of LST anomalies in Istanbul are linked to BCR, surface/volume ratio (SVR), SVF, CGF, and NDVI, with NDVI and BCR having the highest contribution to SUHI formation.	Supervised ML improves urban planning by gauging contribution of urban fabric to microclimate for effective SUHI mitigation.	[60]
CART	SUHI, UTFVI	MODIS, SRTM	an increase in average LST in rural areas compared to the urban core, with a difference of 1-2°C, and a strong negative correlation between LST and NDVI in a range of -0.1 to -0.6.	Uses decadal data for large cities spread across the south Asian region, helps in cross validation & study used remote-sensing indices to provide a comprehensive analysis of the urban thermal environment	[76]
PCA, MLR, SVM	SUHI, LST	Sentinel 2, ASTER	found NDWI, NDBI to be major factors affecting SUHI in Cartago and best LST retrieval from vegetation cover with $R^2=0.78$ & identified areas prone to extreme temperatures with 95% uncertainty.	Compares SVM with NBML algorithm.	[77]

Boosted Regression	LST, Urban form factors	Landsat, Meteorological, DEM	Study on relationship between 2D/3D building patterns and seasonal LST in Hangzhou found dominant impact of MAPA, BCR, FAR, and MAH. Results showed controlling these factors improves thermal environment.	This methodology provides a comprehensive and systematic approach to understanding the effect of 2D/3D architectural pattern on LST and can inform urban planning efforts to mitigate the UHI.	[62]
SOBNNC, MLC, SAM	IS	Sentinel 2A, Landsat8	Overall accuracy of 90.91% and 88.64% and Kappa coefficient of 0.82 and 0.77 for Sentinel-2 and Landsat-8 images respectively.	Considers the contribution of IS to UHIs.	[78]
LR	Heat stress	(CMIP5)	Generalised increase in cooling demand for Africa.	Large scale analysis has been performed for the entire continent of Africa	[79]
	Heat stress	Weather station	Identified substantial variability in max air temperature, heat-related mortality rate & effectively determined high-risk areas using simulated max air temp (MB = 0.11 K, RMSD = 1.40 K).	Combines the heat stress risk assessment and local scale air temperature patterns in a numerical weather prediction model	[80]

XGB	LST	Landsat 8	RMSE of 1.52°C in LST computation	XGBoost with SHAP values accurately measures impact of LULC, LST, and imperviousness on UHI.	[81]
	LST	Landsat 8	M-NDVI in a 150-meter radius had the highest impact on reducing urban LST ($r = -0.63$), while MSI and MNN showed the highest correlation with LST in the 300-1000-meter radius ($r = -0.34$ and 0.34 , respectively).	Study analyses effect of in situ & neighbourhood factors on urban LST in Sari using 200 random pixels & factors like NDBI, %Green, M-NDVI, MPS, MNN, & MSI to determine the most effective approach for alleviating urban LST.	[82]
Markov chain model	LULC	Landsat 5 & 8, SRTM	an increase in built-up areas (+608.8 km ² , 7.5%) & agri land (+285.5 km ² , 3.5%) & decrease in natural vegetation (-831.8 km ² , 10.2%). The trend of built-up increase is expected to continue to 2030, growing 723 km ² (8.9%) & agri land 57.2 km ² (0.7%), causing a decrease in natural vegetation by 663.9 km ² (8.1%).	Methodology of Decision Forest-Markov chain model in LCM tool successfully models future land cover for 2030 with high validation score (0.84) in Total Operating Characteristics (TOC) and higher percentage of components of agreement.	[83]

LUR-Kriging-RF	UHI mitigation	Meteorological station	cold air passages from two of the compensation spaces led to a decrease of 11.88% and 11.82% in PM2.5, and 9.40% and 9.76% in PM10 in the central urban area.	Assesses air cleanliness in compensation spaces and identifies cold air passages through wind direction & PM2.5 & PM10 spatial distributions, providing basis for evaluating impact of urban ventilation corridors on air quality.	[84]
DT, RF	UHI matrix	LST, DEMs, DTMs, LiDAR	approach was developed to identify the correlations between socio-economic and urban morphology features and UHI at a street-level, resulting in five distinct categories of UHI with explicit thresholds and an accuracy of 93%.	Proposed methodology offers an easy-to-use tool for urban planners to investigate UHI impact and demonstrates accurate and consistent results compared to complex models.	[85]
RF	nocturnal UHI, LST, SUHI, URG	Landsat 8 & Meteorological station	predicts nocturnal SUHI in Southern European functional urban areas considering LCZs and daily heat flux components, yielding MSEs less than 0.1 K & emphasizing the importance of LCZ classification in LST prediction.	Study presents energy balance ML approach to predict nocturnal UHI during heatwaves in Southern Europe. Accurate LST predictions emphasized importance of heat flux components and LCZ classification.	[86]

RF, CNN	LST, SUHI	Landsat 8	CNN outperformed RF in mapping LCZ with higher overall accuracy, especially in classifying mixed building and sparse vegetation LCZ classes.	CNN outperforms RF and WUDAPT for mapping LCZs in 4 global cities with improved accuracy for mixed building class.	[87]
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